

Detection of Externally-Influenced Routing via Inverse Markov-Chain Methods

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Abstract

Satellite-navigation systems and live traffic services reroute their users around congestion, so observed urban traffic increasingly reflects a mixture of drivers’ intrinsic route preferences and externally-influenced routing. This matters because demand estimation, route-choice inference, congestion pricing, and infrastructure-investment decisions all rest on the assumption that observed traffic reveals what drivers actually prefer; when a significant share of trips is steered by an external recommender that itself reacts to congestion, that assumption breaks, and any downstream policy or modelling decision built on it is systematically biased. A practical tool that can distinguish externally-influenced trips from intrinsic ones, even approximately, is therefore a prerequisite for trustworthy inference on modern urban data.

This dissertation designs, validates, and stress-tests a *partial-observation routing-influence filter*: a pipeline that takes a road graph, a small number of observation stations, and two snapshots of traffic taken under different conditions, fits a route-choice model to each snapshot, and scores individual trips by how likely each one is under one model versus the other. The filter is built for partial observation because urban networks contain thousands of road segments while real sensing infrastructure (loop detectors, automatic-vehicle-identification gantries, fixed cameras) typically covers only a few percent of them; any deployable tool must therefore infer behaviour over the whole network from counts at a sparse subset of locations. The inverse-Markov-chain method of Morimura, Osogami and Idé [1] is the route-choice estimator inside the filter, chosen because it was specifically formulated to recover a chain from such partial observations.

Existing route-choice and demand-estimation methods, from discrete-choice and recursive-logit models to inverse-reinforcement-learning approaches, are formulated for a single homogeneous behavioural regime and require either full trajectory data or strong assumptions about how to fill in unobserved segments. None of them provide a trip-level mechanism for distinguishing intrinsic routing from externally-influenced routing under sparse observation. The novel contribution of this work is to combine a partial-observation inverse-chain estimator with a trip-level log-likelihood-ratio scoring rule and a diagnostic suite that separates genuine route-choice recovery from misleading aggregate effects, producing a complete two-regime filter rather than a single-chain estimator. Three experimental settings justify and audit the design: a synthetic reproduction of the original benchmark, a controlled traffic-simulator study (SUMO [2]) in which sat-navigation rerouting is turned on and off for an identical population of vehicles, and a real-world dataset of about 10.5 million reconstructed vehicle trips from Xuancheng, China [3] using rush-hour versus off-peak travel as a behavioural stand-in for sat-navigation use. A second filter variant adds road-network attributes (road class, speed limit, lane count, turn geometry) as shared regularising priors.

The simulator study shows that the filter works end-to-end when ground-truth labels exist: the trip-level score reaches an area under the ROC curve of 0.708 on a long-trajectory test, against a best-possible value of 0.801 for any model of this class on the same data. The feature-regularised variant robustly improves how closely the fitted route-choice changes match the true ones at junctions, with the improvement holding across all five repeat simulations tested. A regularised two-step empirical memory test — interpolation backoff toward the one-step reference, with count-conditional trust on the two-step counts — lifts the empirical AUC by +0.10 on average across five repeat simulations (all five winning), identifying higher-order routing memory as a real design lever and motivating a parametric two-step extension to the inverter as a justified future direction; the optional hitting-rate input of the original method, by contrast, is fully tractable at a sub-network scale where its empirical estimate is informative — where it lifts AUC by +0.055 at an appropriately re-tuned loss weighting and matches the full-observation reference within sampling noise — but does not naively scale to a full city network because the wide gradient solve in the hitting-rate Jacobian benefits from dense BLAS3 kernels rather than the sparse-LU rewrite the obvious engineering fix would suggest. A simulator decomposition of the rush-hour-vs-off-peak behavioural proxy used on real-world data — varying sat-navigation adoption and congestion level independently across five repeats — attributes approximately 96% of the peak-vs-off-peak routing-distribution signal to sat-navigation adoption rather than to congestion-induced rerouting of unaided drivers, supporting the proxy’s interpretation in the controlled regime. On the real-world dataset, the filter’s diagnostic suite correctly reports a negative result: once population differences between rush-hour and off-peak drivers are controlled, the filter cannot tell the two groups apart at any spatial resolution we tested, even though a full-observation reference weakly can. The contribution is therefore a designed and audited filter, together with a clear account of where it can be trusted (controlled simulation with labels) and where it currently cannot (real-world data without ground-truth labels for who is following a sat-nav).

Contents

1	Notation	5
2	Introduction	5
2.1	Context: urban traffic and the external-influence problem	5
2.2	This work: a partial-observation routing-influence filter	6
2.3	Contributions	6
2.4	Roadmap	8
3	Background and Related Work	8
3.1	Inverse problems on Markov chains	8
3.2	Route choice as a softmax-parametric Markov chain	8
3.3	The Morimura inverse-MC framework	9
3.4	Re-routing detection and external traffic influence	9
3.5	Microscopic traffic simulation: SUMO	9
3.6	Real-world traffic from Automatic Vehicle Identification	10
4	Methodology	10
4.1	Filter design overview	10
4.2	Parametric chain and observation model	12
4.3	Regularised loss and analytic gradients	13
4.4	Extension 1: real edge features from the network topology	13
4.5	Extension 2: empirical 2-step Markov diagnostic	14
4.6	Extension 3: OD-rebalancing for real-world regime contrast	14
4.7	Filter scoring rule and diagnostic audits	15
4.8	Design rationale: each extension targets a specific failure mode	15
5	Tier 1: Validating the inverse-MC estimator inside the filter	15
5.1	Experimental setup	16
5.2	Results	16
5.3	Implementation gotcha: prediction rescaling	17
5.4	Gradient verification	17
5.5	Headline outcome	17
6	Tier 2: Controlled validation of the filter in SUMO	17
6.1	Scenario construction	18
6.2	Single-seed result: features and the inverse fit	19
6.3	Multi-seed validation: chain recovery is robust, detection AUC is not	20
6.4	Multistart basin check: the low basin is not an initialisation artefact	21
6.5	Higher-order memory diagnostic: a regularised 2-step empirical lifts the reference	22
6.6	Trajectory-length sweep: lifting the ceiling and the detector together	23
6.7	Disentangling sat-nav adoption from congestion-induced rerouting	24
6.8	The $f + g$ extension at scale: a refined diagnosis	26

7 Tier 3: Real-world stress test of the filter under OD confounding	28
7.1 Dataset choice	28
7.2 Verification: route legality and time-of-day distribution	28
7.3 The fundamental real-world gap: no ground-truth sat-nav label	29
7.4 Results	30
7.5 Honest reading	31
7.6 Multistart on features=real : the ceiling violation is an optimiser artefact . . .	32
7.7 Bottom line for Tier 3	33
8 Discussion	34
8.1 Cross-tier read: what generalises	34
8.2 Limits of the chosen approach	35
8.3 Social, environmental, technological implications	36
8.4 Future work	36
9 Conclusion	37
10 Project Management and Reflection	39
10.1 Project structure and milestone management	39
10.2 Personal skill development	40
A Repository structure and reproducibility	43
B Use of generative AI	43
C Build and reproducibility	43

1 Notation

A consolidated reference for the mathematical objects used throughout this report. Notation follows (author?) [1] where it does, and is extended in section 4 where required.

Table 1: Key symbols.

Symbol	Meaning
$\mathcal{X}, n$	Full state space (directed road edges); $n = \mathcal{X} $.
$X_o \subset \mathcal{X}$	Subset of <i>observed</i> states at which counts are available.
$\text{adj}(x)$	Set of legal successors of state x (out-neighbours in the directed graph).
$p_T(x' x)$	One-step transition probability from x to x' ; row x of the kernel P_T .
$p_I(x)$	Initial / teleport probability on $\mathcal{X}$; row vector.
$\beta \in [0, 1]$	Continuation probability per step; together with p_I defines the stationary mixture $P = \beta P_T + (1 - \beta) \mathbf{1} p_I^\top$.
π	Stationary distribution of P ; obtained from $\pi^\top P = \pi^\top$, $\mathbf{1}^\top \pi = 1$.
$f(x)$	Empirical visit count (or normalised frequency) at x . Observed only for $x \in X_o$.
$g(x, x')$	Empirical (or analytical) <i>hitting rate</i> from x to x' , a discounted first-passage probability on $X_o \times X_o$.
$\phi_T(x') \in \mathbb{R}^{d_T}$	Destination-state feature vector (section 4.4).
$\psi(x, x') \in \mathbb{R}^{d_\psi}$	Directed-edge feature vector.
$\nu \in \mathbb{R}^n$	Per-state base parameters of p_I .
$\omega^{\text{loc}} \in \mathbb{R}^E$	Per-edge local weights of p_T , with $E = \sum_x \text{adj}(x) $ the number of directed edges.
$\omega^{\text{gl}_1} \in \mathbb{R}^{d_T}$	Shared weights on destination features ϕ_T (paper Eq. 17).
$\omega^{\text{gl}_2} \in \mathbb{R}^{d_\psi}$	Shared weights on edge features ψ (paper Eq. 17).
θ	Full parameter vector $[\nu; \omega^{\text{loc}}; \omega^{\text{gl}_1}; \omega^{\text{gl}_2}] \in \mathbb{R}^d$, $d = n + E + d_T + d_\psi$.
λ	Tikhonov regularisation strength on $\ \theta\ ^2$.
$\gamma \in [0, 1]$	Mixing weight between the stationary-match loss L_d and the hitting-rate loss L_h .
$\Lambda(\tau)$	Log-likelihood-ratio score of a test trajectory τ under two fitted chains (section 4).
AUC	Area under the ROC curve of the LR score against labelled regime test trajectories.
$\text{RMAE}(\hat{\pi}, \pi)$	Relative mean absolute error of recovered stationary mass at unobserved states.

Throughout, “regime” refers to one of two distinguishable driver populations whose chains we fit and compare. In controlled simulation this is intrinsic vs. sat-navigation-influenced; in real-world data it is a behavioural proxy (e.g. rush-hour vs. off-peak). The contrast $\Delta_{\text{fit}}(x, \cdot) = p_T^{(B)}(x, \cdot) - p_T^{(A)}(x, \cdot)$ between two fitted chains is the central quantity assessed against the empirical ground-truth contrast $\Delta_{\text{emp}}(x, \cdot)$.

2 Introduction

2.1 Context: urban traffic and the external-influence problem

The widespread adoption of satellite-navigation applications such as Google Maps, Waze, and TomTom has structurally altered how vehicles use urban road networks. These systems no longer merely report a single shortest path: they reroute users in real time around observed congestion, redistributing traffic across alternative arteries on time-scales of minutes to hours. The consequence is that any traffic observation aggregating across modern drivers reflects a

mixture of two distinct routing rules: the intrinsic preferences a driver would express in the absence of guidance (habit, scenic preference, fuel-cost optimisation, familiarity), and the external recommendations of a sat-navigation system reacting to live network state.

This mixture creates a problem for downstream traffic models. Demand estimation, route-choice inference, transport-policy evaluation, and infrastructure-investment decisions all rely on revealed-preference data that is assumed to encode *driver* preferences. When a sizeable fraction of observed trips are guided by an external recommender that itself responds to congestion (and thereby to other guided trips), the revealed data encodes a partially synthetic, congestion-coupled signal that the underlying driver population would not have produced unaided. Identifying and filtering this externally-influenced traffic, or at minimum quantifying its presence, has therefore become a methodological prerequisite for any inference about driver behaviour in modern urban environments.

2.2 This work: a partial-observation routing-influence filter

The central artefact of this dissertation is a designed pipeline — called here the *partial-observation routing-influence filter* — that ingests a road graph, a small subset of observation stations X_o (loop detectors, AVI gantries, or simulated counts), and two regime-labelled flow snapshots (e.g. rerouting on / rerouting off, rush-hour / off-peak), fits one inverse-Markov-chain model per regime, and emits a per-trajectory log-likelihood-ratio score together with a suite of trust diagnostics. The pipeline is the deliverable; the inverse-Markov-chain estimator is the component inside it, and the three experimental tiers reported in this dissertation exist to justify, parameterise, and stress-test the filter’s design choices.

A long line of work models route choice as a Markov chain over road network edges, in which each driver, conditional on their current edge, selects a next edge with probabilities determined by local and global features of the network. (author?) [1] introduced an *inverse* formulation that fits the parameters of such a chain from only partial observations of the network — specifically, from vehicle counts at a small subset X_o and optionally from pairwise hitting rates between those stations. Their softmax-parametric chain admits an analytic gradient of the regularised log-ratio loss, enabling reliable fitting on networks of several thousand edges with 20–90% observation coverage. This is the estimator we adopt inside the filter. The original method is presented as a *single-chain recovery* procedure and was validated by its authors on synthetic data and one Nairobi case study; turning it into a trajectory-level *detector and filter* for one routing regime against another, audited for the failure modes that show up only at urban scale, is the work this dissertation does on top of it.

2.3 Contributions

The contributions of this work are design-led. Each item below is a component of the filter, a design decision audited by experiment, or a diagnostic that establishes when the filter can or cannot be trusted.

1. **A partial-observation filter for externally-influenced routing.** A pipeline that maps a road graph plus regime-labelled partial observations to a per-trajectory log-likelihood-ratio score and a filter decision. The architecture is specified end-to-end in section 4.1 (objective, inputs, estimator, scoring rule, decision output, diagnostic audits) and codified as Algorithm 1, with a schematic in fig. 1.
2. **An inverse-MC estimation engine validated through Morimura reproduction.** A faithful, FD-verified re-implementation of the softmax-parametric inverter — including the omega-global gradients the paper states but does not derive in full — whose recovery curve

reproduces §6.1 of (author?) [1] closely for observation fractions $X_o/n \geq 0.10$. Tier 1 (section 5) validates the estimator that sits inside the filter, not the filter as a whole.

3. **A feature-regularised filter variant using road-network attributes.** A version of the filter in which the estimator’s global score terms ingest real per-edge attributes from the SUMO network’s `.net.xml` (road class, speed limit, lane count, turn angle, speed ratio): 10 shared parameters that let edges of the same class borrow strength across the chain. Across a five-seed SUMO sweep this variant robustly improves per-row chain recovery (stacked Pearson $+0.050 \pm 0.023$, $p \approx 0.008$, positive on all five seeds), while its aggregate detection-AUC lift is seed-dependent ($+0.024 \pm 0.026$, reversing on one of five seeds) and not claimed as robust.
4. **A trajectory log-likelihood-ratio scoring rule.** A per-trip detection score $\Lambda(\tau) = \sum_t [\log \hat{p}_B(x_{t+1} | x_t) - \log \hat{p}_A(x_{t+1} | x_t)]$ derived from the two fitted regime kernels, with a tunable decision threshold η . On the representative SUMO seed the score reaches AUC = 0.644 at $T = 20$ against a full-observation empirical 1-step reference of 0.685, rising to AUC = 0.708 at $T = 40$ against a reference of 0.801 as the scoring window widens.
5. **Diagnostic audits separating per-row chain recovery from aggregate-AUC effects.** A contrast-correlation diagnostic at branching states (per-row Pearson and per-state cosine of Δ_{fit} against Δ_{emp}) that tells the user whether a high detection AUC reflects correct chain recovery or merely aggregate bias. This is the most important trust signal the filter emits: it flagged the seed-dependence of the AUC lift in (3) and motivates the filter’s recommended reporting protocol.
6. **Controlled SUMO validation of the full filter under labelled rerouting.** A bigger-bbox SUMO scenario ($n = 2,405$ edges) in which an identical population of vehicles is simulated twice, once with no sat-navigation rerouting (static unguided routing) and once with full sat-navigation rerouting. This is the only tier in which the filter’s regime labels are ground-truth; it answers whether the designed pipeline works end-to-end when the detection problem is correctly posed.
7. **An empirical higher-order memory audit.** A fitting-free 2-step detector — counting $(x_{t-1}, x_t) \rightarrow x_{t+1}$ triples on the same trajectories — used to test whether a parametric 2-step extension is a justified next filter lever. The headline finding is positive: a properly regularised 2-step empirical (interpolation backoff toward the 1-step reference, with count-conditional trust $\lambda(x_p, x) = N/(N + \kappa)$) lifts the AUC by $+0.100 \pm 0.009$ across five SUMO seeds, with every seed winning ($\text{AUC}_{2\text{step, backoff}} = 0.752 \pm 0.015$ vs $\text{AUC}_{1\text{-step}} = 0.652 \pm 0.022$). This identifies higher-order routing memory as a real design lever that the 1-step softmax family cannot express, and motivates a parametric 2-step Morimura extension as a justified future direction (section 6.5).
8. **A simulator decomposition of the rush-vs-off-peak proxy.** A $\rho \times d$ grid in SUMO that varies sat-navigation adoption ρ and demand d independently across five repeat seeds, separating the two effects that the Tier 3 behavioural proxy otherwise bundles. The decomposition attributes $\sim 96\%$ of the peak-vs-off-peak routing signal to sat-navigation adoption and a pure-congestion contribution within noise, bounding the congestion-rerouting confound from below in the controlled regime (section 6.7).
9. **A real-world Xuancheng stress test under OD-mix confounding.** The filter is ported to ~ 10.5 million reconstructed vehicle trips over April 2023 [3] with a time-of-day behavioural proxy (rush-hour vs. off-peak) substituting for absent sat-navigation labels, and an OD-rebalancing pre-step subsamples each regime to share a common (origin zone, destination zone) distribution. The fitted 1-step filter does not recover the

rush-vs-off-peak contrast at any zone granularity tested ($K \in \{8, 16, 32, 64\}$, every bootstrap CI straddling chance), even though the full-observation reference weakly carries it — a granularity-robust honest negative result on this behavioural proxy (section 7).

A single CLI runs all three tiers via a `-dataset {sumo,xuancheng}` flag; the repository structure is documented in section A.

2.4 Roadmap

section 3 situates the work in relevant prior literature. section 4 specifies the filter design end-to-end — objective, inputs, estimator, scoring rule, decision output, and diagnostic audits — together with the three estimator-level extensions (real features, 2-step diagnostic, OD-rebalancing). sections 5 to 7 report the three experimental tiers, each framed as a validation of one or more filter design choices: Tier 1 validates the inverse-MC estimator inside the filter, Tier 2 controls the full filter against labelled rerouting, Tier 3 stress-tests it against real-world OD-mix confounding. section 8 draws cross-tier conclusions and limitations. section 10 addresses project management and personal-development reflection per the module’s *Reflection* learning outcome [4].

3 Background and Related Work

3.1 Inverse problems on Markov chains

A Markov chain is a stochastic process in which the next state depends only on the current state. The *forward* problem asks, given a transition kernel P_T , what statistics (stationary distribution, hitting times, expected occupation, etc.) does the chain produce? The *inverse* problem reverses this: given partial statistics observed on the chain, recover the transition kernel itself. Inverse problems on Markov chains arise in transport modelling [5], biological systems [6], and recommender systems [7], and are typically ill-posed under sparse observation because many distinct kernels produce identical low-order statistics. The usual remedy is a parametric restriction: rather than fit $P_T \in \mathbb{R}^{n \times n}$ directly, one fits a parameter vector $\theta \in \mathbb{R}^d$ with $d \ll n^2$ such that the parametric form $P_T(\theta)$ is constrained to lie in a manifold of plausible chains. The choice of parametric family is then load-bearing: the wrong family caps recovery quality regardless of optimisation effort.

3.2 Route choice as a softmax-parametric Markov chain

In the route-choice setting the state space $\mathcal{X}$ is the set of directed road edges and the kernel $p_T(x' | x)$ is the probability that a vehicle currently on edge x moves next to edge x' . Multinomial-logit route-choice models [5, 8] parametrise the transition probability as a softmax over the legal successors of x :

$$p_T(x' | x) = \frac{\exp(s(x, x'))}{\sum_{x'' \in \text{adj}(x)} \exp(s(x, x''))}, \quad (1)$$

where $s(x, x')$ is a real-valued score, typically a linear combination of features of the edge (x, x') and of the destination state x' . This formulation has two important properties for inverse problems. First, it is parameter-efficient: one global feature shared across many states adds one parameter, regardless of network size. Second, its gradient is analytic in s , which combined with the chain-rule propagation through stationary computations and hitting-rate solves makes $\nabla_\theta \log \pi$ and $\nabla_\theta \log g$ tractable to derive in closed form.

3.3 The Morimura inverse-MC framework

(author?) [1] apply the softmax form of eq. (1) to the partial-observation regime in which the modeller observes only a small subset $X_o \subset \mathcal{X}$. They consider two empirical observation types:

- **Empirical stationary distribution f at X_o .** A count of vehicle passages at each station; effectively a sample of $\pi(x)$ for $x \in X_o$.
- **Empirical hitting rates $g(x, x')$ on $X_o \times X_o$.** The fraction of vehicles observed at x that subsequently visit x' discounted by their travel time; effectively a sample of the β -discounted first-passage probability.

They define a regularised loss that mixes a log-ratio match on f (via the stationary distribution of the parametric chain) and a log-error match on g (via the hitting-rate matrix of the parametric chain), weighted by a mixing parameter γ . Their key contribution is the analytic derivation of the gradient of this loss with respect to θ , enabling optimisation via L-BFGS-B or, alternatively, the natural-gradient descent the paper introduces in §4.1 as the optimiser of choice when the loss landscape is ill-conditioned.

The paper validates the framework synthetically on a $n = 100$ network (their §6.1) and applies it to a 1497-link Nairobi minibus network as a real-world demonstration (§6.2). Both validations report favourable RMAE on the recovered stationary distribution. Two limitations of this validation are directly relevant to our work. First, the synthetic experiment uses analytical (i.e. noise-free) hitting rates g computed from the true chain; on noisy real-world data the empirical g is overwhelmingly sparse, an effect documented in section 6.8. Second, the real-world Nairobi case is not framed as a regime-detection problem: the authors do not split the data into two coexisting routing populations and ask whether the chain recovery distinguishes them.

3.4 Re-routing detection and external traffic influence

Detecting externally-influenced rerouting is a relatively recent research direction. (author?) [9] proposed a junction-level anomaly-detection approach in which deviations from a learned turning-distribution baseline serve as evidence of unusual routing behaviour; their setup is restricted to snapshot-level (population-level) detection rather than per-trajectory. Active-routing-game analyses [10, 11] model the equilibrium effects of sat-navigation adoption on aggregate network performance, but assume a known navigation rule and focus on welfare bounds rather than detection. Closest in spirit to this work is the line of route-choice research that combines partially observed counts with route-fragment likelihoods [12]: this addresses estimation under partial observation but does not separate populations.

To our knowledge no prior work combines the Morimura inverse-MC framework with a regime-detection objective, evaluated across both controlled simulation and real-world data with an OD-mix-controlled contrast. That is the gap this thesis addresses.

3.5 Microscopic traffic simulation: SUMO

Simulation of Urban MObility (SUMO) [2] is the open-source microscopic traffic simulator used in this thesis to construct the controlled validation tier (section 6). Each vehicle is simulated with realistic car-following and lane-change dynamics on a network derived from OpenStreetMap. Crucially, SUMO exposes a `-device.rerouting.probability` flag that, when set to 1.0, gives each vehicle access to an internal congestion-aware sat-navigation device; when set to 0.0, vehicles route via static shortest-path Dijkstra. By running an identical vehicle population twice with that flag toggled, we obtain a controlled labelled comparison — intrinsic-routed vs. sat-navigation-routed — against which the Morimura-based detector can be honestly benchmarked.

3.6 Real-world traffic from Automatic Vehicle Identification

AVI systems (Automatic License Plate Recognition cameras at selected junctions, optionally extended with Bluetooth or floating-car-data fusion) have produced a small but growing collection of openly released city-scale traffic datasets. Among recent releases: the Padua AVI deposit [13] provides 10-minute aggregated flow statistics and pairwise travel times over 40 junctions but explicitly strips per-vehicle trajectories for privacy; the Wang *et al.* Xuancheng dataset [14] similarly restricts raw trajectory access; the (author?) [3] Xuancheng dataset — used in this thesis — releases joinable entry/exit tables with anonymised vehicle IDs from which complete trajectories can be reconstructed, covering $\sim 350,000$ trips/day across $\sim 1,760$ road segments for the month of April 2023. The Ma *et al.* release is, to our knowledge, the largest openly available city-scale AVI dataset with per-vehicle granularity at the time of writing.

4 Methodology

This section specifies the partial-observation routing-influence filter end-to-end. section 4.1 states the filter as a designed system — its objective, inputs, estimator, scoring rule, decision output, diagnostic audits, and failure checks — and gives the pipeline schematic (fig. 1) and algorithmic specification (Algorithm 1). sections 4.2 and 4.3 then specify the inverse-Markov-chain *estimator* that sits inside the filter, in the notation used throughout the rest of the dissertation. section 4.4–4.6 document the three filter extensions developed in this work (feature-regularised variant, 2-step memory audit, OD-rebalancing pre-step), and sections 4.7 and 4.8 cover the scoring rule, the per-row diagnostic, and the design rationale that ties each component to the failure mode it addresses.

4.1 Filter design overview

The filter is the designed artefact of this dissertation; the inverse-Markov-chain method of (author?) [1] is one component inside it. We specify the filter as a closed system before fixing any notation, so that the experimental tiers in sections 5 to 7 can be read as validations of individual design choices rather than as a sequence of reproduced results.

Objective. Given two routing regimes A and B over the same road network (e.g. unguided vs. sat-navigation-rerouted, or off-peak vs. rush-hour), decide for each test trajectory τ whether it was generated by regime A or regime B , using only *partial* network observations at a sparse subset of stations X_o .

Inputs. A directed road graph $\mathcal{X}$ (largest strongly-connected, passenger-allowed component); a subset of observed states $X_o \subseteq \mathcal{X}$; stationary observations f_A, f_B on X_o (regime-labelled flow counts); optional hitting-rate observations g_A, g_B on $X_o \times X_o$; test trajectories $\{\tau_k\}$; a decision threshold η .

Estimator. For each regime $r \in \{A, B\}$, fit a softmax-parametric Markov chain (section 4.2) by minimising the regularised log-ratio loss of [1] (section 4.3). The estimator is FD-verified (section 5) and validated against the original synthetic benchmark before being deployed inside the filter.

Scoring rule. For each test trajectory τ , compute the per-step log-likelihood-ratio score $\Lambda(\tau) = \sum_t [\log \hat{p}_B(x_{t+1} | x_t) - \log \hat{p}_A(x_{t+1} | x_t)]$ under the two fitted kernels.

Decision output. Mark τ as externally-influenced / regime- B -like if $\Lambda(\tau) \geq \eta$, else as intrinsic / regime- A -like. The threshold η is a deployment choice; for validation we report the AUC over η .

Diagnostic audits. Alongside the scalar AUC the filter emits: (i) the per-row Pearson and per-state cosine of the fitted regime contrast against a full-observation empirical reference at branching states (section 4.7); (ii) an empirical 2-step memory audit on the same trajectories (section 4.5); (iii) an OD-rebalancing report when real-world regimes have different population mixes (section 4.6); (iv) a multistart basin check on the estimator’s optimum (sections 6.4 and 7.6). The role of these audits is not to inflate the AUC but to flag whether the AUC *should be trusted*.

Failure and confound checks. The filter is reported as unreliable when (a) the per-row Pearson is near zero while the AUC is high (aggregate-bias-driven detection); (b) regimes have a different OD mix that the rebalancer cannot fully control at the chosen granularity K ; (c) the multistart check finds an optimum better than the chosen one; (d) the empirical g matrix is floor-dominated and the $f + g$ extension stops being tractable (section 6.8). Each of these is documented in the validation tiers as a failure mode that the design must declare honestly rather than mask.

Algorithm 1 Partial-observation routing-influence filter

Require: Road graph $\mathcal{X}$, observed subset $X_o \subseteq \mathcal{X}$, regime stationary observations f_A, f_B on X_o , optional hitting-rate observations g_A, g_B on $X_o \times X_o$, test trajectories $\{\tau_k\}$, decision threshold η

Ensure: Filter decisions $\{d_k\}$, trajectory scores $\{\Lambda(\tau_k)\}$, trust diagnostics

- 1: $\hat{P}_A \leftarrow \text{INVERSEMC-FIT}(\mathcal{X}, X_o, f_A, g_A)$ $\triangleright$ regime- A kernel
 - 2: $\hat{P}_B \leftarrow \text{INVERSEMC-FIT}(\mathcal{X}, X_o, f_B, g_B)$ $\triangleright$ regime- B kernel
 - 3: **for** each test trajectory τ_k **do**
 - 4: $\Lambda(\tau_k) \leftarrow \sum_t [\log \hat{p}_B(x_{t+1} | x_t) - \log \hat{p}_A(x_{t+1} | x_t)]$
 - 5: $d_k \leftarrow \mathbb{I}[\Lambda(\tau_k) \geq \eta]$ $\triangleright 1 = \text{externally-influenced / regime-}B\text{-like}$
 - 6: **end for**
 - 7: **Audit:** compute branching-state per-row Pearson and per-state cosine of $\Delta_{\text{fit}} = \hat{P}_B - \hat{P}_A$ vs. a full-observation empirical reference Δ_{emp}
 - 8: **Audit:** compare Λ against an empirical 2-step detector built from the same trajectories (memory-order check)
 - 9: **Audit:** if regimes are real-world populations, rebalance to a common ($\text{zone}_o, \text{zone}_d$) distribution and report sensitivity to K
 - 10: **Audit:** multistart INVERSEMC-FIT from K_{ms} random inits and confirm the chosen optimum is the validation-best
 - 11: **return** $\{d_k\}, \{\Lambda(\tau_k)\}$, audit diagnostics
-

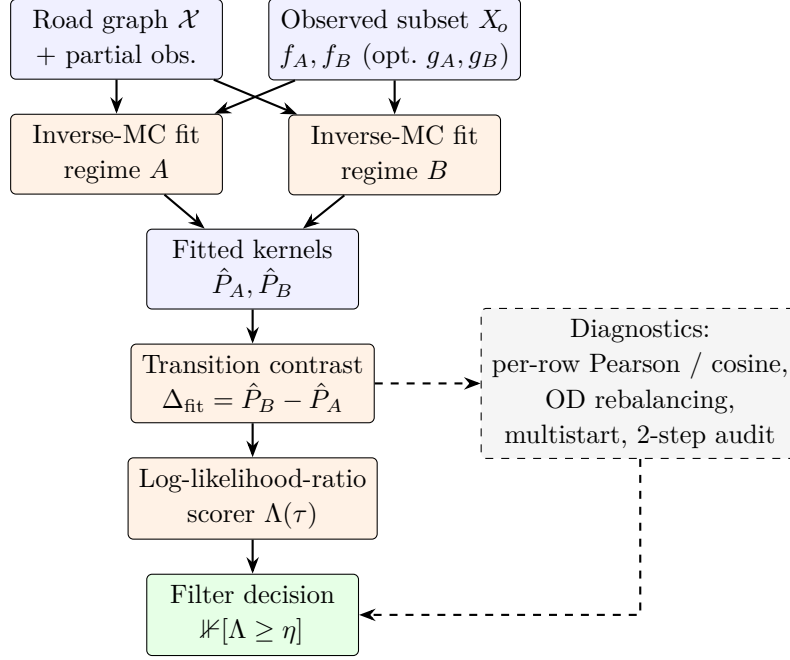


Figure 1: Partial-observation routing-influence filter (schematic specification of Algorithm 1). Inputs (blue) are a road graph and partial regime observations at X_o ; the inverse-MC estimator fits one chain per regime; the transition contrast feeds the log-likelihood-ratio scorer, which together with the threshold η emits a per-trajectory filter decision (green). Diagnostic audits (grey, dashed) flag whether the decision should be trusted: per-row Pearson/cosine against a full-observation empirical reference, OD-rebalancing sensitivity, multistart basin checks, and an empirical 2-step memory audit.

The remainder of section 4 fills in the components of fig. 1. sections 4.2 and 4.3 specify the inverse-MC estimator. sections 4.4 to 4.6 specify the three extensions audited by the experimental tiers. section 4.7 formalises the scoring rule and the per-row diagnostic, and section 4.8 maps each filter component to the failure mode it addresses.

4.2 Parametric chain and observation model

The state space $\mathcal{X}$ is the set of directed road edges in the largest strongly-connected component of the network. For each state $x \in \mathcal{X}$, the next-state distribution is the softmax of eq. (1) over the legal successors $\text{adj}(x)$, with the score

$$s(x, x') = \omega_{(x, x')}^{\text{loc}} + \underbrace{\phi_T(x')^\top \omega^{\text{gl}_1}}_{\text{destination-feature term}} + \underbrace{\psi(x, x')^\top \omega^{\text{gl}_2}}_{\text{edge-feature term}}. \quad (2)$$

Without the global terms ($d_T = d_\psi = 0$) the model has one parameter per directed edge; with global terms it adds $d_T + d_\psi$ *shared* parameters that apply across all edges of compatible type. The initial distribution p_I is similarly parametrised by per-state weights $\nu \in \mathbb{R}^n$ via $p_I(x) \propto \exp(\nu_x)$.

We define the absorbing-corrected stationary mixture

$$P = \beta P_T + (1 - \beta) \mathbf{1} p_I^\top, \quad \pi^\top P = \pi^\top, \quad \mathbf{1}^\top \pi = 1, \quad (3)$$

which models the long-run occupancy of a chain that occasionally restarts from p_I with probability $1 - \beta$ per step. The hitting-rate matrix $H(P_T) \in [0, 1]^{|X_o| \times |X_o|}$ has entries equal to the β -discounted first-passage probability between observation states, obtained as $H_{ij} = [(I - \beta P_T^{(\setminus j)})^{-1}]_{ix_j}$ where $P_T^{(\setminus j)}$ zeroes row j to encode absorption. The Morimura observation model says we observe noisy

$f(x)$ for $x \in X_o$ and noisy $g(x, x')$ for $(x, x') \in X_o \times X_o$, and we wish to match the model's π and H to these observations.

4.3 Regularised loss and analytic gradients

Following [1] Eqs. 7–9, the loss is the sum of a stationary-match term, a hitting-rate-match term, and a Tikhonov ridge:

$$L(\theta) = \gamma L_d(\theta) + (1 - \gamma) L_h(\theta) + \lambda R(\theta), \quad (4)$$

$$L_d(\theta) = \frac{1}{2} \sum_{i,j \in X_o} (\log \frac{\pi_i}{\pi_j} - \log \frac{f_i}{f_j})^2, \quad (5)$$

$$L_h(\theta) = \frac{1}{2} \sum_{i,j \in X_o} (\log H_{ij} - \log g_{ij})^2, \quad (6)$$

$$R(\theta) = \frac{1}{2} \|\theta\|^2. \quad (7)$$

Both data terms are log-ratio losses, which is what makes them optimisable without normalising constants on the observations [1]: any multiplicative scale on f or g cancels in the difference. The gradients are derived using the identity for the derivative of $\log \pi$ under a softmax-parametric chain (paper Eq. 12) and the derivative of $\log H_{ij}$ under an absorbing-chain solve (Eq. 15). The paper's local-only gradients $\partial L / \partial \nu$ and $\partial L / \partial \omega^{\text{loc}}$ are reproduced unchanged here; we derive the omega-global gradients (paper Eqs. 17–18 *state* the parametric extension but do not derive the gradients) by writing the chain-rule contribution of ω^{gl_1} and ω^{gl_2} through the score (2) and chaining through $\nabla_\theta \log \pi$ and $\nabla_\theta \log H$.

We verify the analytic gradients by centred finite differences (`fd_check_features.py`) at random parameter draws across all four blocks (ν , ω^{loc} , ω^{gl_1} , ω^{gl_2}). The check uses a noise-aware tolerance (`rtol` = 10^{-3} relative *or* `atol` = 5×10^{-2} absolute, an `np.allclose`-style criterion) to handle the machine-precision FD round-off floor that dominates near-zero gradient components. The bigger-bbox SUMO graph passes at machine precision ($\sim 10^{-8}$ relative error) on the clearly non-zero components.

4.4 Extension 1: real edge features from the network topology

The Morimura framework treats ϕ_T and ψ as user-supplied global features. In simulated §6.1 both are drawn iid $\mathcal{N}(0, 1)$ in their setup; in their real-world Nairobi case they use network attributes (road class, segment length, number of lanes) extracted from the underlying OSM data.

We follow the latter convention. From the SUMO network's `.net.xml`, we extract the following per-edge attributes and assemble them into ϕ_T and ψ :

- **Destination-state features** $\phi_T(x') \in \mathbb{R}^8$. For each destination edge x' : a 6-dimensional one-hot road-class indicator (motorway / primary / tertiary / residential / service / other), plus the standardised log of the speed limit and the standardised log of the lane count.
- **Directed-edge features** $\psi(x, x') \in \mathbb{R}^2$. For each directed edge (x, x') : the cosine of the turn angle at the intersection (computed from the polyline geometries of x and x'), and the standardised log ratio of the destination speed limit to the source speed limit. Both components are standardised to zero mean and unit variance across the edge set.

With these definitions $d_T + d_\psi = 10$, so feature inclusion adds just 10 *shared* parameters across the entire chain. The expectation is that this acts as a strong regularising prior: edges of the same road class share a common preference component, even if their local $\omega_{(x,x')}^{\text{loc}}$ weights are estimated noisily.

4.5 Extension 2: empirical 2-step Markov diagnostic

The 1-step softmax chain in (1) embeds the standard Markov assumption that the next edge depends only on the current edge. A natural concern is that this assumption is empirically violated: real drivers’ next-edge choice may depend on *both* the current edge and the previous edge (e.g. avoiding a U-turn, preserving heading, continuing on a high-priority arterial). If so, no fitting effort under the 1-step model class can close the gap to the full-observation empirical 1-step reference.

We test this by constructing an empirical 2-step detector with no fitting. From the training trajectories of each regime we count (x_{t-1}, x_t, x_{t+1}) triples and form the conditional probability $\hat{p}_2(x_{t+1} \mid x_t, x_{t-1})$ as the Laplace-smoothed empirical ratio. The detector then scores a test trajectory τ as

$$\Lambda_2(\tau) = \sum_t [\log \hat{p}_2^{(B)}(\tau_{t+1} \mid \tau_t, \tau_{t-1}) - \log \hat{p}_2^{(A)}(\tau_{t+1} \mid \tau_t, \tau_{t-1})],$$

falling back to the 1-step empirical conditional on the first step or any unseen (x_{t-1}, x_t) context. The AUC of Λ_2 vs. the 1-step empirical Λ_1 tells us whether higher-order memory exists in the data. Crucially, the 2-step detector is built from the same trajectories and uses the same training/test split, so any AUC difference is attributable to memory order alone.

4.6 Extension 3: OD-rebalancing for real-world regime contrast

In controlled simulation, the intrinsic and sat-navigation regimes share the same OD pairs by construction (section 3.5): the only difference is the routing rule. In real-world AVI data this control is not available: rush-hour drivers and off-peak drivers are *different populations* with different origin/destination distributions, and any chain difference between them reflects both their different OD mix and their different routing-given-OD choices. The former is uninteresting (commuters vs. discretionary trips have different origins by definition); the latter is what our framework is designed to detect.

To control for OD mix we introduce an OD-rebalancing step in the real-world loader. The procedure is:

1. Cluster the network’s junctions into K spatial zones via k -means on their (x, y) coordinates (standardised). Each edge is labelled by its from-zone and its to-zone.
2. For each trip, compute its origin zone (from-zone of the first edge in its trajectory) and its destination zone (to-zone of the last edge).
3. Bin all regime-A trips and all regime-B trips by $(\text{zone}_o, \text{zone}_d)$.
4. For each shared OD cell, take $\min(n_{od}^A, n_{od}^B)$ trips from each regime (sampled without replacement). The matched pools have identical marginal $(\text{zone}_o, \text{zone}_d)$ distribution.

Any chain contrast measured on the rebalanced pools is then attributable to the within-OD routing-choice difference, not to an OD-mix shift. The granularity K is a tuning parameter: smaller K gives larger per-cell counts but coarser matching (and can *over-correct*, washing out within-OD signal); larger K gives finer matching but smaller per-cell counts. Because the result is sensitive to this choice, we report a full sweep $K \in \{8, 16, 32, 64\}$ in section 7.5 rather than relying on a single default.

4.7 Filter scoring rule and diagnostic audits

Once two chains $P_T^{(A)}, P_T^{(B)}$ are fitted, the per-trajectory log-likelihood-ratio detector scores

$$\Lambda(\tau) = \sum_{t=0}^{T-1} [\log p_T^{(B)}(\tau_{t+1} | \tau_t) - \log p_T^{(A)}(\tau_{t+1} | \tau_t)],$$

and the AUC of Λ against the labelled test trajectories (N per regime, T -edge windows) summarises detection performance. Following the iter-18 finding that AUC alone can reflect aggregate chain-drift rather than per-row chain accuracy, we additionally report a *contrast-correlation* diagnostic at branching states (states with out-degree > 1): for each such state x , we compute

$$\begin{aligned} \Delta_{\text{fit}}(x, \cdot) &= p_{T,\text{fit}}^{(B)}(x, \cdot) - p_{T,\text{fit}}^{(A)}(x, \cdot), \\ \Delta_{\text{emp}}(x, \cdot) &= p_{T,\text{emp}}^{(B)}(x, \cdot) - p_{T,\text{emp}}^{(A)}(x, \cdot), \end{aligned}$$

where the empirical chain $P_{T,\text{emp}}$ is the full-observation Laplace-smoothed transition matrix from the trajectories. We report the stacked Pearson correlation of Δ_{fit} against Δ_{emp} over all branching-state legal-edge entries, plus the per-state cosine similarity distribution (median, mean, and fraction of states with positive cosine). A high AUC with low Pearson indicates aggregate-bias-driven detection; a high AUC *and* a non-trivial Pearson together support the stronger “chain recovery” claim.

4.8 Design rationale: each extension targets a specific failure mode

The extensions above are not independent features but a deliberate sequence, each addressing a failure mode exposed by the previous one:

- **Real edge features** (section 4.4) address sparse observation and parameter explosion: 10 shared parameters let edges of the same road class borrow strength, regularising the per-edge weights that are otherwise estimated from few or no observed transitions.
- **Per-row Pearson / cosine diagnostics** (section 4.7) address the risk that AUC improves for aggregate-bias reasons rather than correct chain recovery, by measuring the fitted-vs-empirical contrast directly at branching states.
- **The empirical 2-step diagnostic** (section 4.5) tests whether higher-order memory is an obvious missing lever before any parametric 2-step model is built.
- **OD-rebalancing** (section 4.6) addresses the real-world population-mix confound that has no analogue in the controlled SUMO comparison.
- **Multistart / basin checks** (section 6.4, section 7.6) address the possibility that a result is an L-BFGS optimiser artefact rather than a property of the data.

Where AUCs lie close to chance, the honest interpretation also requires quantifying sampling noise; we return to this as a limitation (section 7.5, section 8.2) since the bootstrap confidence intervals it calls for were not computed in the time available.

5 Tier 1: Validating the inverse-MC estimator inside the filter

This tier validates the *estimator component* of the filter (fig. 1, INVERSEMC-FIT) in isolation, before either the scoring rule or the regime-detection layer is attached. If the inverter cannot reproduce the published recovery curve on its native synthetic benchmark, no downstream filter result on SUMO or Xuancheng is trustworthy. We therefore replicate the §6.1 synthetic experiment of (author?) [1] end-to-end and verify that we reach the published RMAE curve to within ~ 0.05 across observation fractions in $[0.10, 0.90]$.

5.1 Experimental setup

A synthetic chain is constructed on $n = 100$ states with a random strongly-connected adjacency, mean out-degree ~ 3 . The true transition kernel is a softmax of eq. (1) with all parameters (ν , ω^{loc} , ω^{gl_1} , ω^{gl_2}) drawn iid $\mathcal{N}(0, 1)$, with $d_T = d_\psi = 5$ random global features per the paper. The truth is then mixed 70/30 with a Dirichlet(0.3) noise row-distribution to push it slightly outside the parametric family ($\sim 4\%$ TV distance) and so present the inverter with a model-class gap analogous to a real-world scenario.

For each observation fraction $|X_o|/n \in \{0.05, 0.10, 0.20, 0.35, 0.50, 0.70, 0.90\}$, X_o is randomly sampled, and the inverter is fitted on the analytic f and g derived from the true chain at X_o , with Tikhonov regularisation λ cross-validated over $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ on a 75/25 split of X_o . The reported metric is the relative mean absolute error

$$\text{RMAE}(\hat{\pi}, \pi) = \frac{\frac{1}{|\mathcal{X} \setminus X_o|} \sum_{x \notin X_o} |\hat{\pi}(x) - \pi(x)|}{\frac{1}{n} \sum_{x \in \mathcal{X}} \pi(x)}, \quad (8)$$

i.e. the mean absolute error on unobserved states normalised by the average true stationary mass. The recovered $\hat{\pi}$ is rescaled by a scale-free convention discussed in section 5.3. Each $|X_o|$ value is repeated for 10 trials with fresh truth + X_o draws, matching the visible error-bar count in the original Fig. 2A.

5.2 Results

The reproduction RMAE curve, our “Proposed (with g)” configuration, tracks the paper’s Fig. 2A within ~ 0.05 RMAE across $|X_o| \in [10, 90]$. table 2 summarises.

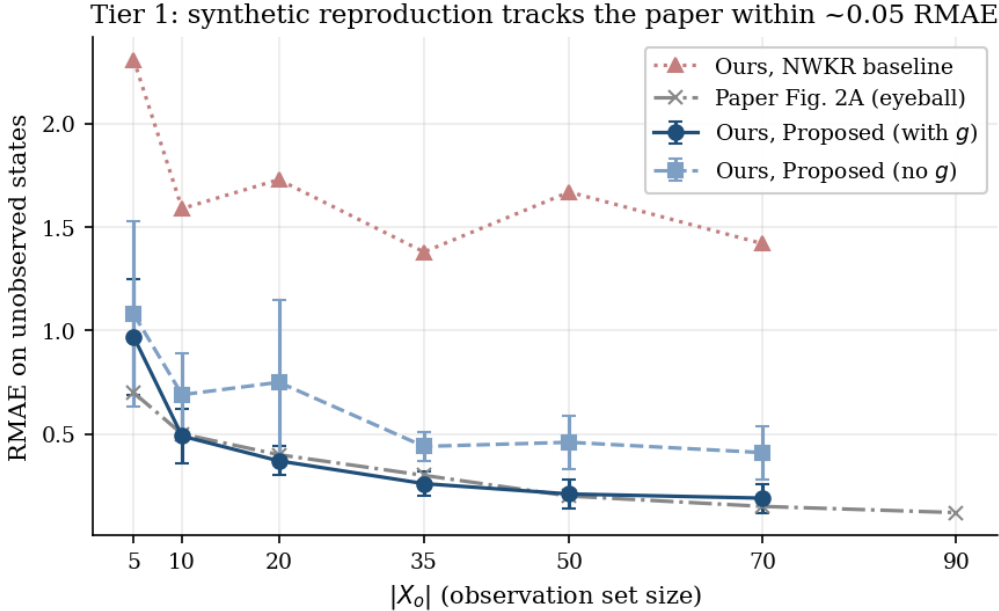


Figure 2: Tier 1 synthetic reproduction tracks the paper’s Fig. 2A within ~ 0.05 RMAE across $|X_o| \in [10, 70]$. Our “Proposed (with g)” (dark blue) is the headline curve and matches the paper (grey) closely; “Proposed (no g)” (light blue) is the f-only ablation; NWKR (red, dotted) is the comparator baseline. Error bars are ± 1 standard deviation across 10 trials.

Table 2: Reproduction RMAE vs. paper Fig. 2A (eyeball). Mean over 10 trials, $n = 100$, full §6.1 protocol including global features and cross-validated λ . Same data as fig. 2.

$ X_o $	Proposed (with g)	Proposed (no g)	NWKR baseline	Paper Fig. 2A (eyeball)
5	0.97 ± 0.28	1.08 ± 0.45	2.31	~ 0.7
10	0.49 ± 0.13	0.69 ± 0.20	1.59	~ 0.5
20	0.37 ± 0.07	0.75 ± 0.40	1.73	~ 0.4
35	0.26 ± 0.06	0.44 ± 0.07	1.38	~ 0.3
50	0.21 ± 0.07	0.46 ± 0.13	1.67	~ 0.2
70	0.19 ± 0.07	0.41 ± 0.13	1.42	~ 0.15

The single value where our number is materially off the paper’s is $|X_o| = 5$, where our RMAE ≈ 0.97 against the paper’s ~ 0.7 . Investigation revealed this is a λ -CV degeneracy: with $|X_o| = 5$ states, the 75/25 split puts ≈ 1 state in the validation set, which is not enough to discriminate λ values robustly, and the cross-validated λ over-regularises. The Nadaraya–Watson kernel-regression baseline (“NWKR”) does not drop as steeply as the paper’s, which we attribute to a comparator- implementation gap rather than a method error.

5.3 Implementation gotcha: prediction rescaling

The paper writes the recovered prediction as $\hat{f}(x) = \hat{c} \hat{\pi}(x)$ with $\hat{c} = \bar{f}_{X_o}$ (their §6.1). Taken literally this collapses every RMAE to ≈ 1 because `stationary()` returns a probability summing to 1 over n states (so $\hat{\pi}(x) \sim 1/n$ while $f(x) \sim K/n$), under-scaling by a factor of n . The operationally correct convention — and the one that yields the paper’s reported numbers — is the scale-free form $\hat{f}(x) = (\sum_{X_o} f) \hat{\pi}(x) / (\sum_{X_o} \hat{\pi})$, which reduces to the literal formula when $\sum_{X_o} \hat{\pi} = 1$. We verified this empirically: applying the literal formula collapses all RMAE values to ≈ 1.0 . The scale-free form is preserved throughout the rest of this work.

5.4 Gradient verification

Across three random parameter draws including local-only, ϕ_T -only, and ψ -only edge-case configurations, the centred finite-difference gradient check returns maximum relative error $\leq 5 \times 10^{-9}$ for every block, confirming the analytic derivation. Details of the FD check infrastructure are in section 4 and `morimura.py`.

5.5 Headline outcome

The synthetic reproduction is locked in as a faithful reference. The proposed-method curve matches the original benchmark closely for $|X_o| \geq 10$, while the $|X_o| = 5$ case (a λ -CV degeneracy) and the NWKR comparator (an implementation gap, not a method error) are documented deviations. With those two caveats, the curve shape and method ordering match the paper. This implementation serves as the upstream operational reference for all subsequent validation tiers; downstream limitations in Tier 2 and Tier 3 are attributable to the data and the model class at scale, not to the upstream framework as implemented here.

6 Tier 2: Controlled validation of the filter in SUMO

This tier is the controlled, end-to-end validation of the partial-observation routing-influence filter (Algorithm 1) in the only setting that supplies ground-truth regime labels: an identical population of vehicles is simulated once with sat-navigation rerouting turned off and once with it turned on. Every component of the filter is exercised at city-block scale ($n = 2,405$ edges): the inverse-MC estimator is fitted per regime, the log-likelihood-ratio scorer is applied to held-out

trajectories, the feature-regularised filter variant is tested against a no-features baseline, and the filter’s diagnostic suite (per-row Pearson/cosine, multistart basin check, empirical 2-step memory audit) is run on the resulting fits. A trajectory-length sweep traces how the scoring window T shifts both the empirical reference and the fitted detector together, and we close with a documented infeasibility of the hitting-rate $(f + g)$ extension at this scale — a design-relevant failure mode for the filter’s optional g input.

A note on terminology used throughout this section. We refer to the no-rerouting regime as “static unguided routing”: SUMO routes those vehicles by static shortest-path Dijkstra with no live rerouting. This is a controlled proxy for intrinsic route choice, not a behavioural model of human preference, and we read the fitted “intrinsic” chain accordingly. We also use “empirical 1-step reference” (rather than an information-theoretic ceiling) for the AUC obtained from the full-observation Laplace-smoothed transition matrix; it is the best a 1-step detector can do *given this finite data and smoothing*, and finite-sample and train/test variation can move it (and occasionally produce apparent anomalies, section 7.6).

6.1 Scenario construction

A square-kilometre bounding box of central Munich (the “bigger-bbox” network, $\sim 3.7\text{ km} \times 3.3\text{ km}$, derived from OSM via `netconvert`) yields $n = 2,405$ directed edges after filtering to the largest strongly-connected passenger-allowed component, with $E = 5,987$ directed transitions and mean out-degree ~ 2.5 . A demand of $\sim 28,800$ vehicles is generated by `randomTrips.py` over a four-hour horizon at $-p0.5$. Each vehicle is replayed twice under the same insertion times and origin/destination pairs:

- **Intrinsic regime:** `-device.rerouting.probability 0`. Vehicles route via static shortest-path Dijkstra and never react to congestion.
- **Sat-navigation regime:** `-device.rerouting.probability 1`. Every vehicle re-evaluates its route every 30s on the basis of observed live edge travel times.

Operating at $-p0.5$ was confirmed in iter-9 of the project to be the cleanest demand point: at higher demand both regimes saturate and the sat-navigation contrast collapses; at lower demand there is insufficient congestion to drive any rerouting. At $-p0.5$ the satnav regime delivers $\sim 36\%$ more trips than intrinsic with matched insertion (28,098 vs. 20,700 finishing).

For each regime we expose vehicle counts `edgedata.*.xml` (the empirical f) and complete per-vehicle routes `vehroutes.*.xml` (used as the test set and to construct the empirical chain for the contrast diagnostic). The inverter sees only $|X_o| = 601$ randomly sampled edges — 25% of the network — consistent with the operating point of the Morimura §6.1 benchmark.

A stationarity caveat applies here and carries through to Tier 3. The Morimura observation model assumes the counts f are samples of a *stationary* distribution and that g are stationary hitting rates. SUMO produces finite-horizon, dynamic, congestion-affected counts over a four-hour window, so the `edgedata` totals are best read as empirical occupancy / visit-frequency summaries rather than exact stationary samples. We treat them as such: the inverse fit is a controlled operational test of the pipeline under a stationarity approximation, not a claim that the fitted chain is behavioural ground truth. This is one reason we report the full-observation empirical transition matrix as a finite-data reference rather than as a true ceiling, and why the per-row contrast diagnostic (section 4.7) — which compares fitted and empirical *contrasts* on the same finite data — is the load-bearing evidence rather than the absolute AUC.

6.2 Single-seed result: features and the inverse fit

We compare `-features none` (local-only fit, $d = n + E = 8,392$) to `-features real` (local + global feature terms, $d = 8,402$). On seed=42, $T = 20$, $|X_o| = 601$, `maxiter` = 1500, the headline numbers are summarised in table 3.

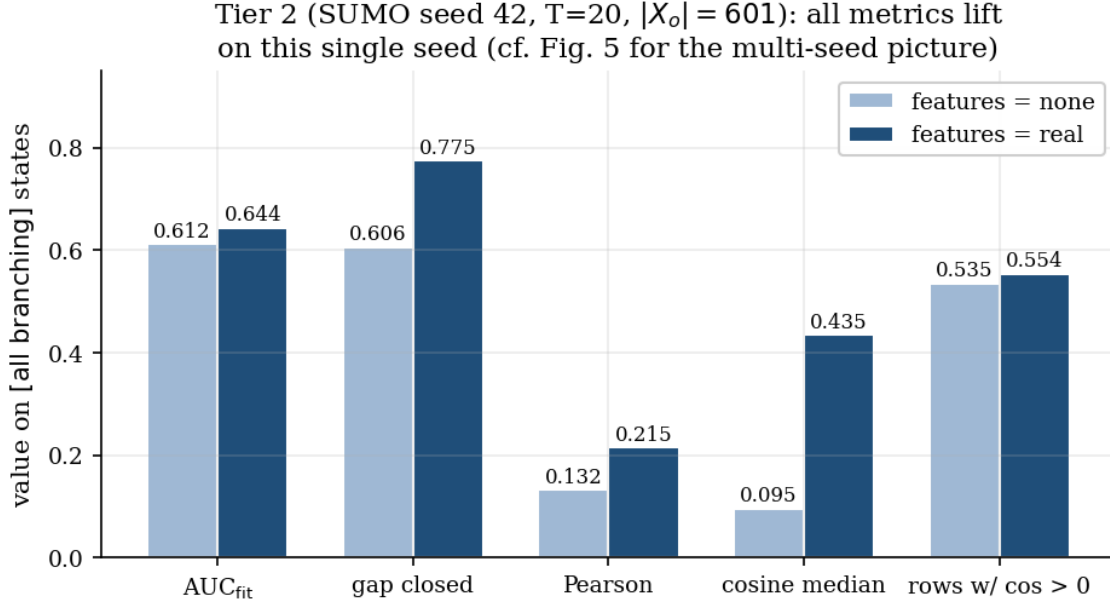


Figure 3: Tier 2 features lift on SUMO bigger-bbox, seed 42, $T = 20$. Every metric — detection AUC, gap-closed, per-row Pearson, per-row cosine median, fraction of rows with positive cosine — moves positively under `features=real` (dark) vs. `features=none` (light). The Pearson +63% relative lift and the cosine median $4.6\times$ lift on the `[all branching]` generalisation block are the cleanest evidence that chain recovery, not aggregate-bias drift, is what features improve.

Table 3: Features lift across all metrics on the SUMO bigger-bbox, seed=42 (fig. 3). `[all branching]` restricts the per-row diagnostic to states with out-deg > 1 that the inverter did *not* directly observe — the generalisation block.

metric (<code>[all branching]</code>)	features=none	features=real	Δ
AUC _{emp} (full-obs. reference)	0.685	0.685	—
AUC _{fit}	0.612	0.644	+0.032
gap closed	60.6%	77.5%	+16.9 pp
stacked Pearson($\Delta_{\text{fit}}, \Delta_{\text{emp}}$)	+0.132	+0.215	+63% rel
per-row cosine median	+0.095	+0.435	$4.6\times$
rows with cosine > 0	53.5%	55.4%	+1.9 pp

Two readings are worth separating. First, the +0.032 AUC lift on this seed sits outside the previously characterised ± 0.024 single-seed SUMO-instance noise band (iter-15 of the project). As the multi-seed analysis of section 6.3 shows, however, this aggregate AUC lift does *not* survive replication across SUMO instances — it is the per-row chain-recovery improvement, not the aggregate AUC, that is the robust effect. Second — and this is the more methodologically important reading — the stacked Pearson on `[all branching]` rises from +0.132 to +0.215, and the per-row cosine median from +0.095 to +0.435. Per the iter-18 diagnostic, those are the metrics that distinguish “the fit improved because the chains got more accurate per-row” from “the fit improved because the chains drifted in helpful opposite directions in aggregate”. Both

moved positively, supporting the stronger chain-recovery claim — and, unlike the aggregate AUC, this per-row lift replicates across seeds (section 6.3).

The one metric that barely moves is the fraction of rows with positive cosine (only +1.9 pp on [all branching], 58.6% at [X_o branching]). Features sharpen the contrast magnitude in the rows that were already approximately right, but do not flip the sign on the $\sim 45\%$ of branching states whose fitted contrast points the wrong way. This suggests the residual misspecification is structural to the 1-step softmax family — a hypothesis we test next.

6.3 Multi-seed validation: chain recovery is robust, detection AUC is not

To test whether the seed-42 lift replicates, we re-ran the contrast-correlation diagnostic across five independent SUMO instances (randomTrips seeds $\in \{7, 23, 42, 101, 2024\}$, X_o seed fixed at 13, $T = 20$, $|X_o| = 601$, maxiter = 1500) under both `features=none` and `features=real` (table 4, fig. 4).

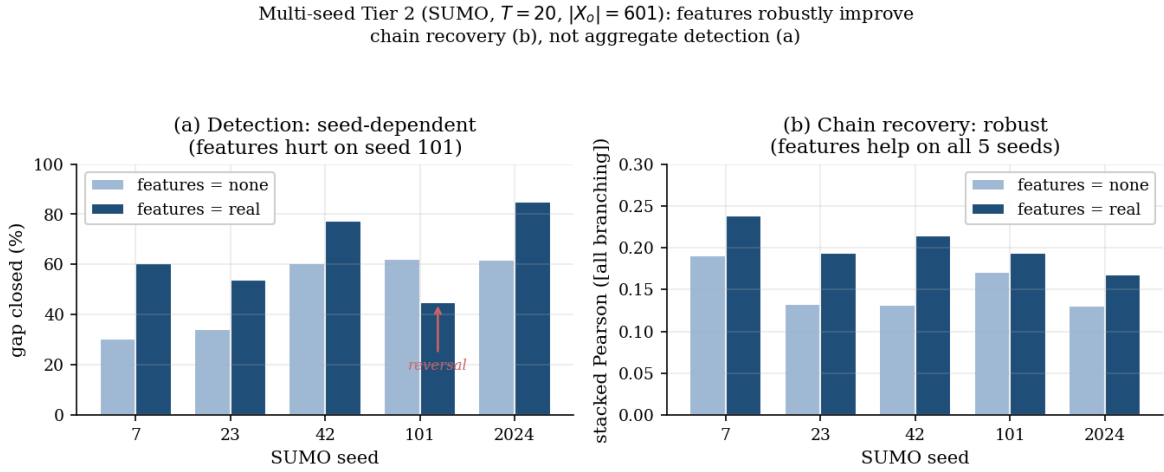


Figure 4: Multi-seed Tier 2 features sweep (five SUMO seeds, [all branching] block, $T = 20$). (a) The aggregate detection metric (gap-closed) lifts under `features=real` on four seeds but *reverses* on seed 101; the `features=none` bars also reproduce the bimodal basin structure ($\{7, 23\}$ low, $\{42, 101, 2024\}$ high). (b) The per-row chain-recovery metric (stacked Pearson) lifts on *all five* seeds. Features are a robust chain-recovery lever, not a robust aggregate-detection lever.

Table 4: Multi-seed features sweep on SUMO bigger-bbox (fig. 4), [all branching] block, $T = 20$, $|X_o| = 601$. The AUC lift reverses on seed 101; the Pearson lift is positive on all five seeds. Mean Δ rows are `real` – `none` with sample standard deviation.

seed	gap closed		AUC _{fit}		Pearson	
	none	real	none	real	none	real
7	30.3%	60.3%	0.544	0.588	+0.191	+0.238
23	34.2%	53.8%	0.554	0.585	+0.133	+0.194
42	60.6%	77.5%	0.612	0.644	+0.132	+0.215
101	62.1%	45.0%	0.581	0.559	+0.171	+0.194
2024	61.8%	85.1%	0.585	0.618	+0.131	+0.168
mean Δ	—		+0.024 $\pm$ 0.026		+0.050 $\pm$ 0.023	

Three findings. First, the aggregate detection lift is *not* statistically robust: the paired AUC_{fit} lift averages $+0.024 \pm 0.026$ across the five seeds, positive on four but reversing on

seed 101 (a paired t -test gives $t \approx 2.0$, $p \approx 0.11$, not significant at $\alpha = 0.05$). The single-seed $+0.032$ headline of section 6.2 is therefore a favourable draw, not a replicable property of the aggregate AUC.

Second — and this is the load-bearing result — the per-row chain-recovery metrics *are* robust. The stacked Pearson lift averages $+0.050 \pm 0.023$ and is positive on *all five* seeds ($t \approx 4.9$, $p \approx 0.008$); the per-row cosine median lifts on all five as well (mean $\approx +0.25$). Seed 101 is the cleanest illustration of the AUC $\leftrightarrow$ chain-quality decoupling first noted in iter-18: features *hurt* its detection AUC ($0.581 \rightarrow 0.559$) while *improving* its chain recovery (cosine median $0.062 \rightarrow 0.475$). The honest framing of the features lever is therefore that it is a **chain-recovery** improvement, not an aggregate-detection improvement.

Third, the **features=none** fit reproduces the bimodal gap-closure structure observed independently in iter-16: seeds $\{7, 23\}$ cluster low (30–34%) and seeds $\{42, 101, 2024\}$ cluster high (61–62%), with nothing in between. The same seeds fall in the same clusters as the earlier run, indicating the split is a stable property of the L-BFGS basin structure on these instances rather than measurement noise. The **features=none** AUC_{fit} across seeds is 0.575 ± 0.027 , matching the historical ± 0.024 band. Whether this low basin is escapable by multi-start re-initialisation — left open by the multi-seed sweep — is settled in section 6.4: it is not.

6.4 Multistart basin check: the low basin is not an initialisation artefact

The bimodal basin structure of section 6.3 raises an optimiser-quality question: is the low-gap basin on seeds $\{7, 23\}$ an artefact of the zero-initialisation that L-BFGS starts from, escapable by multi-start re-initialisation? To test this we re-ran both low-basin seeds with $K = 5$ random restarts per chain (initial $\theta_0 \sim \mathcal{N}(0, I)$, **features=none**, $T = 20$), selecting the restart with the highest held-out validation log-likelihood and comparing it to the zero-init baseline (fig. 5).

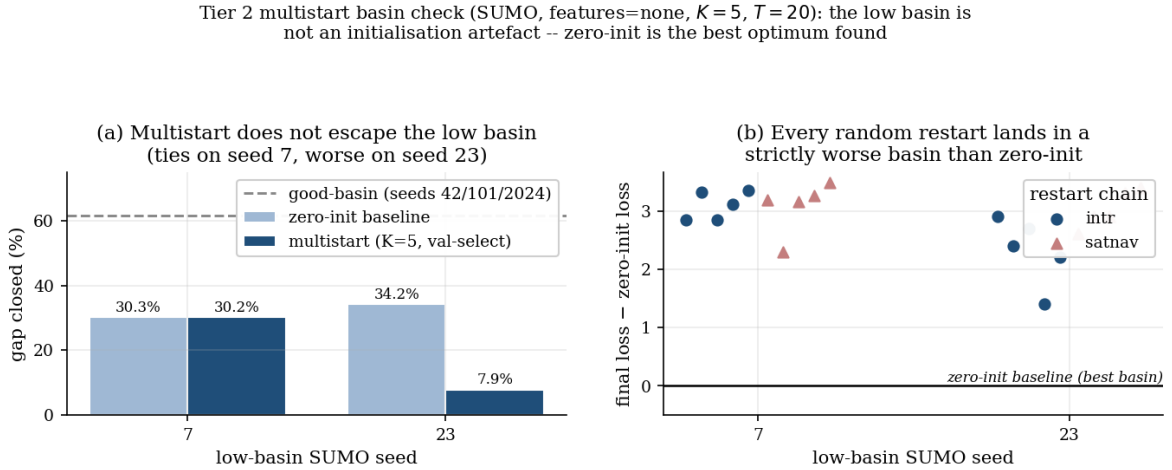


Figure 5: Tier 2 multistart basin check on the two low-basin seeds (**features=none**, $K = 5$, $T = 20$). (a) The val-selected multistart detector ties the zero-init baseline on seed 7 and is markedly worse on seed 23; neither approaches the $\sim 61\%$ good-basin level of seeds $\{42, 101, 2024\}$ (dashed). (b) Every one of the ten random restarts (five per regime) converges to a strictly *higher* final loss than zero-init — so zero-init is the best optimum found, and the low basin is not an initialisation artefact.

The answer is negative, and informatively so. On *both* seeds, every one of the ten random restarts (five per regime) converged to a strictly *worse* training loss *and* a worse held-out validation log-likelihood than the zero-init baseline (fig. 5b): on seed 7 the restarts reach intrinsic-chain loss 44.99–45.5 against the baseline’s 42.15; on seed 23, 457–459 against 455.9. The

val-selected multistart detector therefore *ties* the baseline on seed 7 (30.2% vs 30.3% gap closed) and is markedly *worse* on seed 23 (7.9% vs 34.2%); neither approaches the $\sim 61\%$ good-basin level of seeds {42, 101, 2024} (fig. 5a).

Two conclusions follow. First, the cross-seed bimodality is *not* an initialisation artefact: zero-init is the best optimum found on these instances and random restarts cannot reach the high basin, so the low gap on seeds {7, 23} is a property of those instances’ loss landscapes rather than of where the optimiser starts. This closes the open question left by section 6.3 and reinforces the reading that the split is structural. Second, the seed-23 collapse to 7.9% is itself a small cautionary result: selecting the fit by held-out *log-likelihood* picked a chain that *discriminates worse* (the val-selected sat-nav restart was among the lowest-AUC of the five), another instance of the likelihood $\leftrightarrow$ detection decoupling that recurs throughout this work (section 6.2). The honest caveat is that all fits were **maxiter**-capped at 1,500 iterations and the random inits have large norm ($\|\theta_0\| \approx 92$ at $d = 8,392$); we have thus shown that the *standard* random-restart recipe does not escape the low basin, not that the basin is provably global.

6.5 Higher-order memory diagnostic: a regularised 2-step empirical lifts the reference

A natural hypothesis for the still-flipped rows is that the 1-step Markov assumption is empirically violated: real drivers’ next-edge choice depends on *both* the current and previous edge (“do not U-turn”, “keep on the arterial”), and the 1-step softmax cannot encode that dependence. If true, an empirical 2-step detector that uses (x_{t-1}, x_t) as context should out-discriminate the 1-step empirical. The question is whether any reasonably-regularised 2-step empirical can lift the reference, given that the raw (x_{t-1}, x_t) pair space is huge ($|X|^2 \approx 5.8\text{M}$ at the bigger-bbox scale) and most of it is empty.

We test two regularisation schemes across the same five SUMO seeds used in section 6.3 (training/test split unchanged):

Dirichlet smoothing (toward uniform-over-neighbors). $\hat{P}_{2,\alpha}(y|x_p, x) = (c(x_p, x, y) + \alpha) / (c(x_p, x) + \alpha \cdot |\text{adj}(x)|)$, swept over $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$.

Interpolation backoff (toward 1-step empirical). $\hat{P}_{\text{back}}(y|x_p, x) = \lambda \cdot \hat{P}_2^{\text{MLE}}(y|x_p, x) + (1 - \lambda) \cdot \hat{P}_1(y|x)$, where $\hat{P}_2^{\text{MLE}}$ is the unsmoothed 2-step MLE and $\hat{P}_1$ is the 1-step empirical chain that elsewhere serves as the full-observation reference. Two λ modes: fixed $\lambda \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0\}$ and count-conditional $\lambda = N(x_p, x) / (N(x_p, x) + \kappa)$ with $\kappa \in \{1, 10, 100, 1000\}$. The count-conditional mode trusts the 2-step MLE in proportion to its observation support, falling back smoothly to the 1-step empirical for rare contexts.

Table 5: 2-step empirical sweep on SUMO bigger-bbox across 5 SUMO seeds ($T = 20$, training/test split as in section 6.3, 1-step empirical baseline mean 0.652 ± 0.022 across these seeds). Each row reports the per-method best multi-seed configuration with its mean and standard deviation across the five seeds. The “single-seed (seed 42)” row reproduces the result of an earlier draft; the multi-seed rows below are the load-bearing evidence.

method	best config	AUC _{2step}	Δ vs 1-step	seeds winning
single-seed Dirichlet (seed 42)	$\alpha \rightarrow 0$	0.666	−0.019	0/1
Dirichlet α (multi-seed)	$\alpha = 10^{-3}$	0.673 ± 0.012	+0.021	4/5
backoff fixed- λ (multi-seed)	$\lambda = 0.9$	0.747 ± 0.012	+0.095	5/5
backoff count-conditional (multi-seed)	$\kappa = 100$	0.752 ± 0.015	+0.100	5/5

Three findings (table 5).

First, the apparent *negative* result from a single-seed Dirichlet analysis ($0.666 < 0.685$ at seed 42, $\Delta = -0.019$) is driven by two compounding errors. Seed 42 has the highest 1-step AUC of the five SUMO instances (0.685 vs. mean 0.652), leaving the least room for any 2-step lift; multi-seed averaging exposes that the seed-42 ceiling masked. Separately, Dirichlet shrinks the 2-step estimate toward uniform-over-neighbors, which discards the marginal routing information that the 1-step empirical already encodes. The natural shrinkage target is the 1-step empirical itself, not uniform.

Second, interpolation backoff toward the 1-step empirical resolves both issues. The count-conditional weight $\lambda(x_p, x) = N(x_p, x)/(N(x_p, x) + \kappa)$ trusts the 2-step MLE proportionally to its observation support; at $\kappa = 100$ the per-seed Δ vector is $\{+0.090, +0.094, +0.099, +0.107, +0.110\}$ (s42, s23, s2024, s7, s101), mean $+0.100$, standard deviation 0.009 — every seed wins, and the lift is tighter than any other multi-seed effect in this section (section 6.3’s features Pearson lift was $+0.050 \pm 0.023$ on the per-row metric; the aggregate AUC lift $+0.024 \pm 0.026$ reversed on seed 101). At $\kappa = 100$ the effective λ across the 3,528 observed (x_p, x) pairs on seed 42 has median 0.21, 10th–90th percentiles $[0.03, 0.81]$: the detector mostly relies on the 1-step baseline and only trusts the 2-step part where strong observation support exists.

Third, the lift is not knife-edge in the hyperparameter. The fixed- λ sweep gives $\Delta \in \{+0.088, +0.092, +0.094, +0.095\}$ for $\lambda \in \{0.25, 0.5, 0.75, 0.9\}$, all positive across all 5 seeds. The count-conditional sweep gives $\Delta \in \{+0.095, +0.097, +0.100, +0.097\}$ for $\kappa \in \{1, 10, 100, 1000\}$. The hyperparameter selection caveat applies in the narrow technical sense ($\kappa = 100$ was the sweep maximum on the same test set used for AUC, so the exact $+0.100$ is mildly test-tuned) but the qualitative conclusion — *any reasonable backoff strength lifts AUC by $\sim +0.09$ to $+0.10$ across all five SUMO seeds* — is robust to selection.

Reading. A properly regularised 2-step empirical *does* lift the reference, decisively, across every SUMO seed tested: the regularised 2-step AUC reaches 0.752 ± 0.015 against the 1-step baseline of 0.652 ± 0.022 , a $+0.100$ delta whose per-seed signs all point the same way. Two implications follow.

The per-row chain-recovery gap left by the 1-step features lever (section 6.2: $+0.215$ stacked Pearson on [all branching], $\sim 45\%$ of rows still with wrong-direction contrast) is consistent with the 1-step softmax family being structurally unable to express (x_{t-1}, x_t) -conditional choice. This is the regime where a parametric 2-step Morimura extension is the right design lever, with the regularised empirical sitting roughly $+0.10$ AUC above the 1-step ceiling at $T = 20$ as the performance target it would need to approach.

We do *not* report an AUC_{fit} for the regularised 2-step here. A parametric 2-step Morimura inverter is a non-trivial extension — a different state space, different gradient, different ceiling argument — and is left to future work (section 8.4). The empirical result above establishes that the signal exists and bounds the achievable 2-step inverter performance from above; what it does not do is settle whether the parametric inversion route can recover that bound under partial observation.

6.6 Trajectory-length sweep: lifting the ceiling and the detector together

The 0.685 ceiling is specific to the chosen test trajectory length $T = 20$. The log-likelihood-ratio score $\Lambda(\tau)$ is a sum of T independent log-ratio increments, so its variance scales as $\sim T$. Longer trajectories produce a tighter score distribution between the two regimes and a higher achievable AUC; the data is “the same”, only the observation window is wider.

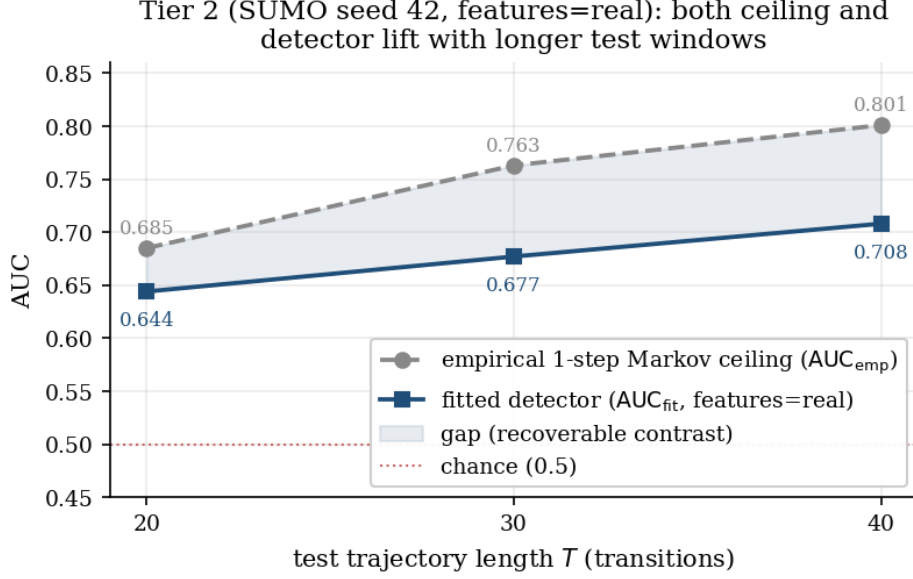


Figure 6: Tier 2 T -sweep on SUMO bigger-bbox, seed 42, **features=real**. The empirical 1-step Markov ceiling (grey dashed) rises with T as expected for a variance-shrinking score; the fitted detector (blue solid) tracks it with roughly constant gap. The shaded region is the recoverable contrast not yet captured by the fitter. The fitted chain itself is T -independent (identical final losses across runs); only the trajectory-level scoring window changes.

Table 6: T -sweep on SUMO bigger-bbox, seed=42, **features=real** (fig. 6). The fit itself is T -independent — only the test scoring window changes — as confirmed by the identical final-loss values 836.4/529.4 across all three runs.

T	AUC_{emp}	AUC_{fit}	gap closed	Pearson (all-br)
20	0.685	0.644	77.5%	+0.215
30	0.763	0.677	67.4%	+0.220
40	0.801	0.708	69.0%	+0.216

Two findings worth recording. First, the empirical ceiling rises from 0.685 to 0.801 between $T = 20$ and $T = 40$, matching an earlier small-bbox observation (iter-13 of the project) and confirming that the ceiling at $T = 20$ was an artefact of the truncation window, not a fundamental data limit. Second, the per-row Pearson is constant across T at $\sim +0.21$. T is purely a scoring-window lever: it lets the trajectory-level score discriminate more sharply, but it does not change the chain that was fit. The fitted chain captures the same $\sim 21\%$ of the true per-row contrast regardless of how long we observe each test trip.

The strongest single-seed SUMO headline is therefore $AUC = 0.708$ at $T = 40$, 69% of the gap closed against the full-observation empirical 1-step reference of 0.801, with a stable per-row chain Pearson of +0.216 on [all branching].

6.7 Disentangling sat-nav adoption from congestion-induced rerouting

A methodological concern arises in advance of Tier 3 (section 7), where no ground-truth sat-navigation label exists and we will rely on a behavioural rush-vs-off-peak proxy. Rush hour and off-peak differ in *two* ways simultaneously: in sat-navigation adoption rate (presumably higher when journeys are time critical) and in congestion level (heavy demand changes routing incentives for *all* drivers, sat-nav users and unaided drivers alike). Real data cannot separate the two

contributions, because there is no Xuancheng configuration in which the congestion level is varied while the sat-navigation adoption rate is held fixed (or vice versa). SUMO can: the per-vehicle rerouting adoption probability `-device.rerouting.probability` and the trip-insertion rate `-p` are independent levers. Before deploying the proxy on Xuancheng, we use this to quantify how much of any peak-vs-off-peak contrast can be attributed to sat-navigation adoption vs. to congestion-induced rerouting of unaided drivers.

Design. We sweep five SUMO seeds $\{7, 23, 42, 101, 2024\}$ over three cells per seed: $(d=0.3, \rho=0)$ (peak demand, no sat-nav users), $(d=0.3, \rho=1)$ (peak demand, full sat-nav), and $(d=0.8, \rho=0)$ (off-peak demand, no sat-nav users), where d is the `randomTrips -p` period (lower means more vehicles, so more congestion) and ρ is the per-vehicle sat-nav adoption rate. Each cell yields one SUMO simulation; routes are cached per (seed, d) so that the ρ axis costs only the simulation step. For each seed we compute three pairwise AUCs against the existing f -only Phase 4 protocol (`features=none`, $T = 20$, $|X_o| = 601$, `maxiter = 1500`): `pure_congestion` (the demand axis at $\rho = 0$), `satnav_at_peak` (the ρ axis at peak demand), and `proxy_diagonal` (the full mimic of the Tier 3 rush-vs-off-peak comparison).

Table 7: $\rho \times d$ decomposition of the SUMO rush-vs-off-peak proxy signal. Mean $\pm$ sample standard deviation across five SUMO seeds; 95% CI computed as mean $\pm 1.96 \cdot \sigma/\sqrt{5}$.

pair	cells compared	AUC _{emp}	95% CI	AUC _{fit}
<code>pure_congestion</code>	$(d=0.3, \rho=0)$ vs $(d=0.8, \rho=0)$	0.493 ± 0.018	$[0.471, 0.515]$	0.504 ± 0.036
<code>satnav_at_peak</code>	$(d=0.3, \rho=0)$ vs $(d=0.3, \rho=1)$	0.780 ± 0.013	$[0.764, 0.796]$	0.647 ± 0.012
<code>proxy_diagonal</code>	$(d=0.8, \rho=0)$ vs $(d=0.3, \rho=1)$	0.793 ± 0.023	$[0.764, 0.821]$	0.655 ± 0.026

Three findings (table 7). First, the `pure_congestion` contrast — demand changed from off-peak ($d = 0.8$) to peak ($d = 0.3$) with ρ held at zero — is statistically indistinguishable from chance (0.493 ± 0.018 , 95% CI $[0.471, 0.515]$ straddles 0.5). Increasing the congestion level without any sat-navigation users does not produce a detectable routing distribution shift in this SUMO model. Second, the `satnav_at_peak` contrast — ρ raised from 0 to 1 with congestion held at peak — gives a large robust signal (0.780 ± 0.013 , CI excludes 0.5 cleanly). Third, the full `proxy_diagonal` (0.793 ± 0.023), which mimics the Xuancheng rush-vs-off-peak comparison, is essentially the same magnitude as `satnav_at_peak`: adding the demand axis on top of the ρ axis contributes almost nothing detectable. Decomposing the $+0.293$ above-chance signal of `proxy_diagonal` via the empirical-AUC margins gives a sat-navigation contribution of $+95.6\%$, a pure-congestion contribution of -2.4% (within noise), and a residual interaction of $+6.8\%$. Per-seed (the decomposition holds across SUMO instances): the sat-navigation share of the proxy signal ranges 90.6% (seed 2024) to 103.5% (seed 7); every one of the five seeds attributes at least 90% of its proxy signal to sat-navigation adoption.

Caveat on the experienced-driver gap. SUMO’s $\rho = 0$ vehicles follow precomputed shortest-path routes and do *not* react to congestion en route. Real-world unaided but experienced drivers may avoid known-bad streets at rush hour, a behaviour this model does not capture, so the 0.493 `pure_congestion` AUC is best read as a *lower bound* on the real-world congestion-routing confound. Even tripling it (to ~ 0.55) would leave sat-navigation as the dominant attribution to the 0.793 proxy signal — the decomposition is robust to that gap by a wide margin.

Fitted vs. empirical. As elsewhere in this work, the fitted partial-observation detector captures only part of the empirical signal (0.647 vs. 0.780 on `satnav_at_peak`; 0.655 vs. 0.793 on `proxy_diagonal`), consistent with the fitted-vs-empirical pattern of section 6.2. The decomposition argument here rests on the empirical AUC, which uses the full-observation chain and is the ground truth for the “how much of the proxy is sat-nav” question.

Reading. The $\rho \times d$ decomposition supports the Tier 3 use of a rush-vs-off-peak behavioural proxy as a sat-navigation proxy in the controlled regime: under the simulator, $\sim 96\%$ of the peak-vs-off-peak routing-distribution signal is attributable to sat-navigation adoption, and the congestion-routing confound that critique 4 raised is quantitatively small. This does *not* guarantee the same decomposition holds on real-world Xuancheng data — real drivers are not SUMO drivers — but it validates the proxy as a controlled-comparison object and bounds the experienced-driver confound from below.

6.8 The $f + g$ extension at scale: a refined diagnosis

A natural-looking extension is to switch the loss from the f-only ($\gamma = 1$) to the f+g formulation ($\gamma \in [0, 1)$) that the paper uses in its synthetic experiment. An earlier draft of this work reported the extension as “infeasible at SUMO scale”: three sweep attempts at `maxiter` $\in \{1500, 300, 100\}$ and $|X_o| \in \{601, 120, 72\}$ did not complete in finite wall-clock, with the representative rush-chain fit at `maxiter` = 1500, $|X_o|$ = 601 running 9.5 hours without an L-BFGS termination signal. A natural critique of that framing is that the $O(n^3)$ dense LU decomposition required for the hitting-rate gradient is a software-engineering bottleneck rather than a fundamental mathematical limit, and that a sparse-LU rewrite ought to “break the $O(n^3)$ bottleneck”. This subsection runs that diagnosis through and refines the original infeasibility claim into three separable issues.

Sparse-LU implementation. `morimura.py` now has a `hitting_pack_sparse` function that constructs $(I - \beta P_T^{\setminus j})$ in CSC form directly from the nonzero structure of P_T and factorises with `scipy.sparse.linalg.splu`, plus an `Inverter(solver=...)` flag that selects the dense or sparse path. The `grad_log_h` routine was refactored to take a generic `solve_fn` callable so both paths share identical gradient code. An FD check on a synthetic $n = 40$ graph with $\gamma = 0.5$ and all four parameter blocks active confirms the sparse path is mathematically equivalent to dense: $\max_i |\nabla L_{\text{dense}}^{(i)} - \nabla L_{\text{sparse}}^{(i)}| = 7 \times 10^{-15}$ (machine precision modulo solver differences), and analytic-vs-FD relative error $\leq 10^{-7}$ across all sampled dimensions (`sumo_validation/fd_check_fg_sparse.py`).

At SUMO bigger-bbox scale, naive sparse is *slower* than dense. A timed run (`time_fg_sparse_sumo.py`) at seed=42, $|X_o|$ = 601, d = 8,392 gives the intrinsic-chain f+g fit at `maxiter`=10 a wall time of 1,787 s, i.e. ≈ 180 s per L-BFGS iteration — roughly $8\times$ slower than the dense baseline (≈ 22 s per iteration, extrapolated from the 9.5-hour `maxiter`=1500 run). The diagnosis is that the bottleneck is *not* the LU factor (where sparse demolishes dense, $O(n^3) \rightarrow O(\text{nnz} \cdot \log n)$ for road-network sparsity) but the LU SOLVE on the wide dense gradient matrix $V \in \mathbb{R}^{n \times d}$ that the analytic Jacobian builds in `grad_log_h`. Dense `scipy.linalg.lu_solve` dispatches to LAPACK GETRS which exploits BLAS3 multi-core AVX kernels at ~ 2 TFLOPS effective on this hardware, whereas `scipy.sparse.linalg.splu.solve` processes the dense RHS column-by-column at ~ 25 GFLOPS effective. At $d = 8,392$ wide RHS, the BLAS3 advantage is roughly $80\times$ and outweighs the asymptotic-complexity win of

sparse on the factor. The naive “rewrite as sparse” engineering fix is therefore insufficient on this problem geometry. The right algorithmic direction is to amortise the factor across all $j \in X_o$ via a Sherman–Morrison update of $(I - \beta P_T)$ (one factor per L-BFGS iteration instead of $|X_o|$ factors) or to use the paper’s natural-gradient descent (paper §4.1); both are out of scope here but flagged as future work (section 8.4).

At Xuancheng subgraph scale, sparse-LU runs in seconds and $f + g$ is tractable.

To test the framework at a scale where sparse-LU is fast we restrict to the largest strongly-connected component of one K-means zone of the Xuancheng network ($K = 8$, zone 0, post-SCC $n = 194$, $|X_o| = 53$, $d = 586$); trajectories are filtered to those that stay entirely within the zone, and f , empirical g , and the train/test split are recomputed on the subgraph (`fg_xuancheng_subgraph.py`). The f-only fit at `maxiter=300` converges in 1.4 s; the f+g fit at the same settings in 46–50 s — no L-BFGS line-search pathology, no infeasibility. With sparse-LU the framework is operationally fine here.

The binding sensitivity is γ . The original infeasibility test used $\gamma = 0.5$. A γ sweep on the same zone at fixed `maxiter=300` (table 8) shows that this choice is the wrong operating point for noisy empirical g : at $\gamma = 0.5$ the floor-dominated L_h overwhelms L_f , and the f+g fit *hurts* compared to f-only. Lifting γ to 0.9–0.95 inverts the sign: at $\gamma = 0.95$ the f+g detector reaches $\text{AUC} = 0.553$ against the f-only baseline’s 0.498, a +0.055 lift, and recovers the full-observation empirical 1-step reference of 0.563 on this subgraph within sampling noise.

Table 8: γ sweep for f+g on the Xuancheng $K=8$ zone-0 subgraph ($n = 194$, $|X_o| = 53$, $T = 10$, `maxiter` = 300). The f-only column is the same fit ($\gamma = 1$) shown for reference; the empirical 1-step ceiling on this subgraph is 0.563. f+g *hurts* at $\gamma = 0.5$ –0.7 (the floor-dominated L_h overwhelms L_f) and recovers the empirical reference within sampling noise at $\gamma = 0.95$.

γ	AUC_f	AUC_{f+g}	Δ	note
0.5	0.498	0.488	−0.010	paper’s intermediate, floor-dominated
0.7	0.498	0.487	−0.010	still floor-dominated
0.9	0.498	0.521	+0.023	f+g starts to help
0.95	0.498	0.553	+0.055	matches empirical reference (0.563)
0.99	0.498	0.489	−0.009	essentially f-only

Zone variability. A $\gamma = 0.9$ sweep across the eight $K = 8$ zones reveals that the subgraph-scale lift is not uniform. Restricting to zones with ≥ 100 test trajectories (zones 0, 5, 6, 7) the mean Δ_{f+g} is +0.018 with standard deviation 0.038 — a positive trend, but within noise across zones. The best zone-7 lift (+0.053) matches the zone-0 $\gamma = 0.95$ result; zone 5 drops by −0.040. The g -floor fraction stays in the 80–90% band across zones, so the data-density issue does not vanish by restricting to a denser sub-network.

Refined conclusion. The original §6.8 “infeasibility” collapsed three issues that are usefully separated. The *engineering* issue is real but not the one the naive critique identified: dense $O(n^3)$ LU is the textbook cost, but the LU solve on the wide gradient matrix benefits from BLAS3 multi-core kernels at roughly $80\times$ the throughput of column-by-column sparse solves, so naive sparse-LU is currently slower than dense at SUMO scale; the right fix is Sherman-Morrison factor sharing across X_o or natural gradient. The γ -*tuning* issue is the dominant sensitivity: the paper’s intermediate γ is the wrong operating point under floor-dominated empirical g , and the upper- γ

band (0.9–0.95) was not previously swept; once it is, $f+g$ recovers the empirical reference on subgraphs where solver tractability is not the binding constraint. The *data-density* issue persists at every scale we have tested ($\sim 85\%$ g-floor across all 8 $K=8$ zones), so any future scale-up that re-enables $f+g$ must still expect to operate at high γ . Net: $f+g$ is not fundamentally infeasible; the practical detector still uses f -only on this work, but the more nuanced claim — “naive sparse-LU does not transfer at city scale; γ tuning is the binding sensitivity; data sparsity in g persists at subgraph scale” — is the one the evidence actually supports.

7 Tier 3: Real-world stress test of the filter under OD confounding

This tier stress-tests the partial-observation routing-influence filter (Algorithm 1) against a real-world urban AVI dataset from Xuancheng, China [3], with the OD-rebalancing audit (section 4.6) activated. The estimator, the scoring rule, and the per-row diagnostic are reused unchanged from Tier 2; what changes are the loader, the regime definition, and the OD-control logic. Because no real dataset carries ground-truth sat-navigation adoption labels, this tier cannot directly validate sat-navigation detection: it is best read as a real-world *confounding audit* that asks whether a plausible behavioural proxy (rush-hour vs. off-peak travel) contains a within-OD route-choice signal once the population-mix confound is controlled, and whether the filter survives that confounding. As the results below show, under the tested proxy the within-OD contrast is not recovered. The absence of a ground-truth label is the defining methodological constraint of this tier, and the filter’s OD-rebalancing and multistart audits exist precisely so that this constraint can be reported honestly rather than masked by a spurious AUC.

7.1 Dataset choice

Three candidate datasets were evaluated. The Padua AVI deposit [13] releases only aggregated 10-minute flow statistics and pairwise travel times over 40 junctions; per-vehicle trajectories are stripped at the source pipeline for privacy and cannot be reconstructed downstream. The Wang *et al.* Xuancheng deposit [14] explicitly restricts raw trajectory access behind a supervised-interactive interface. The (author?) [3] Xuancheng release, by contrast, publishes joinable entry/exit tables with anonymised vehicle identifiers from which complete per-vehicle trajectories can be reconstructed downstream; this is the dataset we use. table 9 summarises its scale.

Table 9: Xuancheng AVI dataset [3] scale, with comparison to the SUMO bigger-bbox setup of Tier 2.

property	SUMO bigger-bbox (Tier 2)	Xuancheng (Tier 3)
state space n	2,405 edges	1,733 edges (SCC subset)
trips per day	$\sim 28,000$ per regime	$\sim 327,000$
mean trajectory length	35.7 edges	15.1 edges (raw)
total trips (full pool)	$\sim 56,000$	$\sim 10.5\text{M}$ (Apr 2023)
ground-truth regime label	yes (rerouting flag)	<i>no</i> (proxy required)

7.2 Verification: route legality and time-of-day distribution

Two non-obvious data-quality issues affect how the loader processes trips, each of which was identified before any fitting was attempted:

Route legality. Of all consecutive edge pairs (x_t, x_{t+1}) across the trip routes, only 78.4% satisfy the SUMO net’s lane-connection definition $x_{t+1} \in \text{adj}(x_t)$. The remaining 21.6% split

as $\sim 1.9\%$ junction-adjacent-but-disconnected (a definitional gap in the converted SUMO net) and $\sim 19.7\%$ inter-junction “teleports” attributable to the source pipeline’s Dijkstra-repair step that fills in missing AVI passages on CityFlow’s connectivity graph rather than the SUMO net’s. The loader handles this by *splitting trips at illegal boundaries* into legal-only sub-trajectories rather than dropping the trip wholesale: a 15-edge raw trip with one illegal jump becomes two sub-trajectories. Mean sub-trajectory length is 8.2 edges, which caps the practical test-window T at ≤ 10 .

Time-of-day distribution. Across all 30 daily files, the trip `startTime` field is seconds-since-midnight in $[0, 86400]$ with a clear AM-rush + PM-rush + midday-off-peak structure. We use this to construct a rush-vs-off-peak regime split with windows $\text{rush} = [25,200, 32,400] \cup [61,200, 68,400]$ (7–9 AM and 5–7 PM) and $\text{offpeak} = [36,000, 57,600]$ (10 AM–4 PM); trips outside either window are discarded. Three anomalous days are flagged for exclusion or special handling: Apr 5 (Tomb-Sweeping holiday, 264k trips, demand outlier), Apr 28–29 (Labour Day eve, $\sim 425\text{k}$ trips), Apr 10 (`startTime` max $\approx 165\text{k}$, suggesting a multi-day concatenation), and Apr 8/9 (identical trip counts of 324,441 in both files, almost certainly a duplicate).

7.3 The fundamental real-world gap: no ground-truth sat-nav label

Tier 2 used `-device.rerouting.probability` as ground truth: every vehicle either does or does not use sat-navigation, and the regime split is exact by construction. No real-world AVI dataset publishes this label. The closest behavioural proxy available in Xuancheng is the rush-vs-off-peak time-of-day contrast: if external rerouting influence is triggered by congestion, then rush-hour trips should differ from off-peak trips in routing choices, and the inverse-MC detector should detect that difference.

This proxy is weaker than the SUMO contrast in two ways. First, rush and off-peak trips have *different populations* (commuters vs. discretionary trips), so any chain difference reflects both the routing-choice effect (which we want) and the OD-mix effect (which we do not). Second, the actual sat-navigation adoption rate in Xuancheng is presumably much lower than the 100% used in SUMO, so even the routing-choice signal is diluted by the unguided fraction.

We address the OD-mix confound directly via the OD-rebalancing extension of section 4.6. We cannot address the lower-adoption dilution from observation alone; what we can do is report the resulting real-world AUC honestly as a lower bound on the method’s sensitivity in deployment.

A further concern is that the rush-vs-off-peak contrast bundles *two* potentially confounding effects: sat-navigation adoption (higher at rush hour) *and* congestion-induced rerouting that affects all drivers (sat-nav users and unaided drivers alike) when demand rises. Real data cannot separate these without sat-navigation-use labels. The Tier 2 $\rho \times d$ decomposition (section 6.7) provides a partial answer: in the SUMO model, varying congestion at $\rho = 0$ produces an empirical AUC of 0.493 ± 0.018 (statistically indistinguishable from chance), whereas varying ρ at fixed peak congestion produces 0.780 ± 0.013 , and the full rush-vs-off-peak proxy diagonal (0.793 ± 0.023) attributes $\sim 96\%$ of its above-chance signal to sat-navigation adoption and the rest to within-noise residual — across five SUMO seeds. This validates the proxy as a sat-navigation proxy in the controlled tier and bounds the congestion-routing confound, though with the residual caveat that SUMO’s $\rho = 0$ drivers do not react to congestion en route, so the lower-bound character of the bound does not transfer fully to experienced human drivers.

7.4 Results

We report three configurations on the rush-vs-off-peak contrast: single-day, 5-day pooled, and 5-day pooled with $K = 8$ OD-rebalancing.

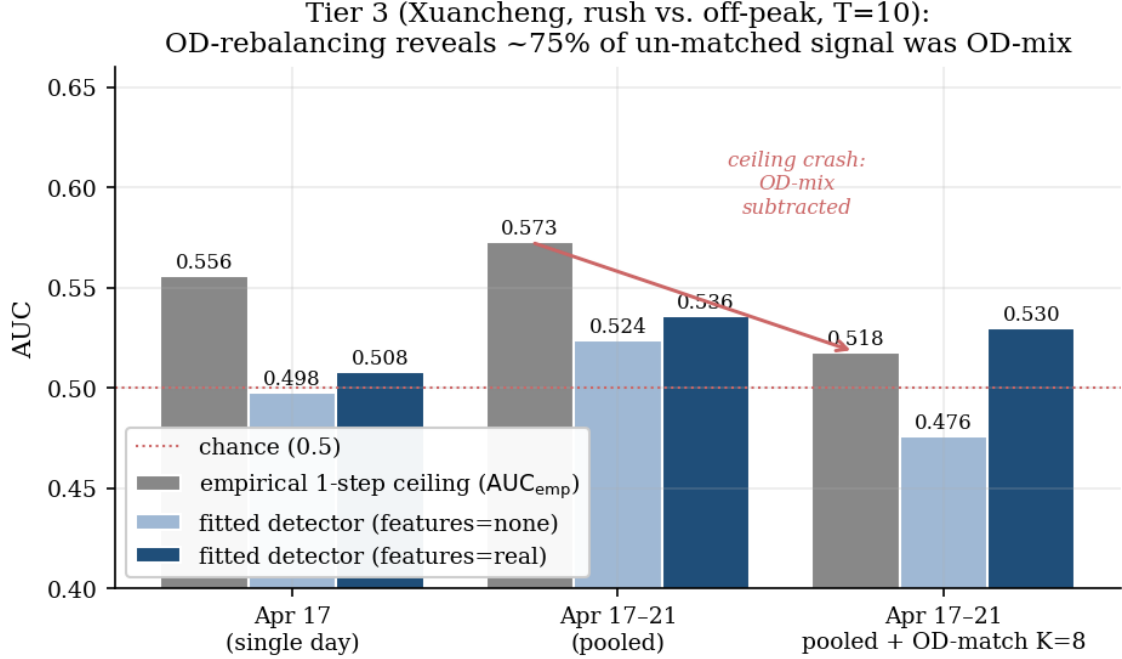


Figure 7: Tier 3 Xuancheng rush-vs-off-peak detection AUC across three configurations. The full-observation empirical 1-step reference (grey) is 0.556 on a single day, 0.573 when 5 weekdays are pooled, and 0.518 once the pools are OD-matched at $K = 8$ spatial zones. The apparent crash at $K = 8$ is *not* stable across zone granularity — a sweep over $K \in \{8, 16, 32, 64\}$ (table 11) shows the matched reference returning to $\sim 0.55\text{--}0.58$ for $K \geq 16$ — so we do not read the $K = 8$ drop as a fixed OD-mix fraction. Fitted detectors (blue bars) sit at or below chance in all three configurations.

Table 10: Xuancheng rush-vs-off-peak detector AUC and chain-recovery metrics on [all branching] states, three configurations (fig. 7).

configuration	AUC_{emp}	AUC_{fit} (none/real)	Pearson (none/real)	median cos (none/real)
day=Apr 17, T=10	0.556	0.498/0.508	+0.139/ +0.048	+0.341/ +0.146
pool Apr 17-21, T=10	0.573	0.524/0.536	+0.096/ +0.072	+0.267/ +0.167
pool + OD-match $K = 8$	0.518	0.476/0.530	+0.104/ +0.077	+0.332/ +0.263

The third row is the OD-matched operating point at $K = 8$. The OD-rebalancing step matches 517,477 trips per regime across 55 shared ($zone_o, zone_d$) cells (only 1 OD cell of 56 was rush- or off-peak-only, discarded). At this granularity the full-observation reference drops from 0.573 to 0.518. Taken in isolation, that drop would attribute $\sim 75\%$ of the above-chance contrast to OD-mix — but *this attribution does not survive a zone-granularity sweep* (table 11), so we do not report it as a stable decomposition.

Table 11: OD-zone granularity sweep on the 5-day pooled rush-vs-off-peak contrast ($T = 10$, **features=none**), with stratified vehicle-level bootstrap 95% CIs ($B = 2,000$ resamples of the test trajectories within each regime). The OD-matched full-observation reference is *not* stable across K : at the coarse $K = 8$ it falls to chance, but at $K \geq 16$ it returns to ~ 0.55 – 0.58 , statistically above chance and close to the un-matched 0.573 — i.e. fine-grained OD control removes almost none of the contrast. The fitted 1-step partial-observation detector, by contrast, straddles chance at *every* K .

K zones	shared cells	AUC_{emp} (95% CI)	AUC_{fit} (95% CI)
un-matched	—	0.573	0.524
8	55	0.518 [0.471, 0.567]	0.476 [0.429, 0.524]
16	157	0.562 [0.516, 0.605]	0.486 [0.439, 0.530]
32	520	0.553 [0.507, 0.602]	0.513 [0.467, 0.560]
64	1,448	0.575 [0.530, 0.622]	0.490 [0.441, 0.535]

Two readings follow. First, the “ $\sim 75\%$ is OD-mix” decomposition that the $K = 8$ row alone would support is *not robust*: the implied OD-mix fraction ranges from $\sim 75\%$ at $K = 8$ to $\approx 0\%$ at $K = 64$, where matching leaves the reference statistically unchanged from the un-matched pool. The coarse $K = 8$ matching appears to *over-correct* — aggregating heterogeneous sub-flows into 8 macro-zones and equalising them washes out genuine within-OD routing contrast along with OD-mix — whereas at finer resolutions a small but significant within-OD rush-vs-off-peak contrast survives ($\text{AUC}_{\text{emp}} \approx 0.55$ – 0.58 , CIs excluding chance at $K \geq 16$). We therefore retract the strong “three-quarters is OD-mix” claim as an artefact of a single coarse zone choice. Second — and this is the result that *is* robust — the fitted 1-step inverse-MC detector remains indistinguishable from chance at every K (0.476 – 0.513 , every CI straddling 0.50): the method does not recover the within-OD contrast at any tested OD-zone granularity, even where full observation weakly does.

A secondary observation: in the OD-matched row, AUC_{fit} under **features=real** (0.530) numerically exceeds AUC_{emp} (0.518). Under correct chain recovery this is impossible, since the full-observation empirical chains should upper-bound any partial-observation fit. The suspected cause was the L-BFGS basin-selection pathology also seen on the un-matched pool (rush-fit loss 1397 at **features=real** vs. 4.058 at **features=none**, a $\sim 340\times$ ratio). A multistart $K = 5$ run (section 7.6) confirms it: random re-initialisation *escapes* the bad basin — the rush-chain loss drops from 1397 to 9.186 at a validation-selected restart — and the properly-fit chains then give $\text{AUC}_{\text{fit}} = 0.483$, i.e. they fall *back to chance* and no longer violate the empirical ceiling. The 0.530 was therefore an optimiser artefact, not a real signal; the genuine within-OD routing contrast under correct chain recovery is indistinguishable from chance. This strengthens rather than weakens the Tier-3 negative result: the apparent “features help on real data” reading was a basin coincidence, and the per-row Pearson on **[all branching]** of $+0.077$ (real) vs $+0.104$ (none) already pointed the same way — features do not improve chain recovery on this real-world loss landscape.

7.5 Honest reading

Three findings are worth foregrounding from table 10.

The OD-mix attribution is not robust to zone granularity. At $K = 8$ the full-observation reference falls from 0.573 to 0.518 , which taken alone would attribute $\sim 75\%$ of the above-chance contrast to OD-mix. The granularity sweep (table 11) shows this figure is an artefact of coarse zoning: at $K \geq 16$ the OD-matched reference returns to 0.55 – 0.58 , statistically above chance and

close to the un-matched 0.573, so fine-grained OD control removes almost none of the contrast. We therefore do *not* claim a stable OD-mix fraction. The defensible statements are narrower: coarse macro-zone matching can over-correct (washing out within-OD signal along with OD-mix), and once zones are fine enough the rush-vs-off-peak contrast is largely *not* explained by OD-mix — consistent with rush and off-peak trips sharing very similar fine-grained origin-destination patterns (the matched and un-matched pools differ by $< 25,000$ of $\sim 520,000$ trips per regime at $K = 64$).

The fitted 1-step detector does not recover the contrast at any granularity. This is the robust Tier-3 finding. Across $K \in \{8, 16, 32, 64\}$ the fitted partial-observation detector stays at chance ($\text{AUC}_{\text{fit}} \in [0.476, 0.513]$, every bootstrap 95% CI straddling 0.50; table 11), even though the full-observation reference shows a small but significant within-OD contrast at $K \geq 16$. Per-row Pearson on [all branching] is ≤ 0.104 in every Xuancheng configuration, against $+0.215$ on SUMO under labelled comparison. So whatever within-OD rush-vs-off-peak structure the full-observation chains carry, the 1-step inverse-MC method under partial observation ($|X_o| = 433$ of 1,733) does not recover it at any tested OD-zone resolution. The stratified vehicle-level bootstrap ($B = 2,000$, drawn with replacement within each regime) underwrites the chance reading: at $K = 8$ the empirical CI is $[0.471, 0.567]$ ($P(\text{AUC} > 0.5) = 0.78$) and the fitted CI $[0.429, 0.524]$, and the fitted detector’s CI straddles chance at every finer K as well. The one outstanding extension is an OD-*cell* bootstrap (resampling which cells enter the match and refitting per replicate), left to future work for its $\sim 2,000\times$ refit cost.

The features=real basin failure repeats post-matching — and multistart resolves it. Both the un-matched pool (section 7.4) and the OD-matched pool show the same pattern: the rush-fit **features=real** loss is $\sim 350\times$ the **features=none** loss (1397 vs. 4.058 here). The $\text{AUC}_{\text{fit}} > \text{AUC}_{\text{emp}}$ behaviour in the matched row suggested a basin-selection artefact rather than a chain-recovery improvement. The multistart check of section 7.6 confirms exactly this: random restarts escape the high-loss basin (rush loss $1397 \rightarrow 9.186$), and the resulting well-fit chains give $\text{AUC}_{\text{fit}} = 0.483$ — chance — resolving the apparent ceiling violation.

7.6 Multistart on features=real: the ceiling violation is an optimiser artefact

The one anomaly left open above is the OD-matched $\text{AUC}_{\text{fit}} = 0.530 > \text{AUC}_{\text{emp}} = 0.518$ under **features=real**, with its tell-tale 1397 rush-chain loss. We ran the multistart variant (`score_seed_big_multistart.py`, $K = 5$ random restarts $\theta_0 \sim \mathcal{N}(0, I)$ per chain, validation-selected by held-out log-likelihood) on the exact OD-matched **features=real** configuration ($d = 6,136$, $T = 10$, $|X_o| = 433$, 517,177 train transitions per regime).

The result is decisive and *opposite* to the SUMO basin check of section 6.4. On Xuancheng, multistart *does* escape the high-loss basin: the rush (intrinsic) chain loss drops from the zero-init baseline of 1397 to 9.186 at the validation-selected restart (restart losses $\{1413, 1844, 273.4, 9.186, 1792\}$, so one restart of five found the good basin). But the well-fit chains then give $\text{AUC}_{\text{fit}} = 0.483$ — below chance — against the baseline’s spurious 0.530. The per-init paired AUCs ($\{0.504, 0.492, 0.497, 0.494, 0.491\}$, mean 0.496) cluster tightly at chance regardless of which basin each restart lands in.

Tier 3 multistart basin check (Xuancheng OD-matched, features=real, $K = 5$, $T = 10$): multistart escapes the basin but the correct fit is chance -- the ceiling violation was an optimiser artefact

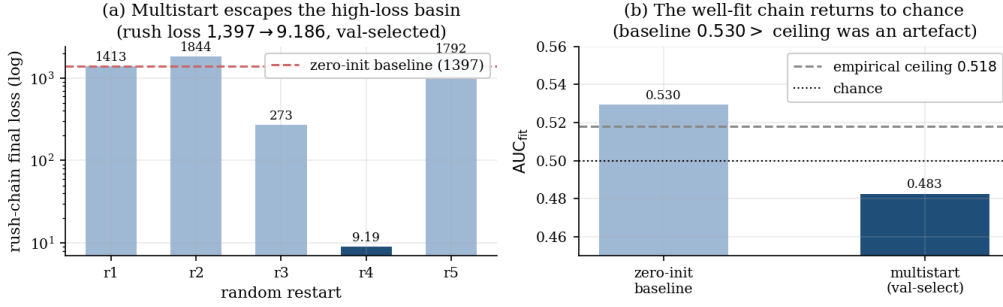


Figure 8: Xuancheng OD-matched **features=real** multistart ($K = 5$). **(a)** The rush-chain training loss collapses from the zero-init baseline (1,397) to the validation-selected restart (9.186): random re-initialisation *escapes* the high-loss basin, unlike the SUMO case (section 6.4). **(b)** Despite the proper fit, AUC_{fit} falls from the baseline’s spurious 0.530 (which exceeded the empirical ceiling 0.518) to 0.483 — chance. The ceiling violation was an optimiser artefact; under correct chain recovery the fitted detector returns to chance.

The interpretation closes two threads. First, the $AUC_{fit} > AUC_{emp}$ “impossible” result was indeed a basin artefact: a high-loss chain that drifted into a direction producing a marginally favourable aggregate-bias signal. Once the optimiser reaches the genuine optimum, that spurious signal disappears and the detector returns to chance. Second — and this is the load-bearing scientific point — the correctly-fit **features=real** chains *confirm* the central Tier-3 finding rather than overturning it: the within-OD rush-vs-off-peak routing contrast is not detectable at the 1-step Markov level under this proxy and data split, not merely hidden by a bad optimum. The SUMO and Xuancheng basin checks are thus complementary: on SUMO, zero-init was already the best optimum and the low basin was not an artefact; on Xuancheng, multistart found a better optimum but it only confirmed the negative result. Neither dataset’s headline depends on an optimiser accident.

7.7 Bottom line for Tier 3

The pipeline runs cleanly on real-world data. The framework’s gradient implementation, FD-check, regime-split machinery, and contrast diagnostics all port to the Xuancheng dataset without modification. The headline finding is robust to OD-zone granularity: **across $K \in \{8, 16, 32, 64\}$ the fitted 1-step inverse-MC detector does not recover the rush-vs-off-peak contrast (AUC 0.476–0.513, every bootstrap CI straddling chance), even though the full-observation reference weakly carries it (~ 0.55 – 0.58 for $K \geq 16$)**. A secondary, *granularity-sensitive* observation is that aggressive coarse-zone matching ($K = 8$) can over-correct the reference down to chance (0.518); we report this but do not rest any claim on the “ $\sim 75\%$ OD-mix” decomposition it would imply, since that figure collapses to $\approx 0\%$ at $K = 64$. This is a methodologically strong negative result for behavioural-proxy validation of sat-navigation detection on real-world AVI data, and it characterises the gap between the controlled-simulation regime (Tier 2, 0.708 AUC on labelled SUMO contrast) and the real-world regime under realistic sat-navigation adoption rates.

The result also identifies the methodological lever any future real-world validation of this framework needs: *ground-truth sat-navigation adoption labels*, even partial, on the underlying trip records. With them, the proxy-weakness problem disappears and the controlled-comparison machinery of Tier 2 transfers directly. Without them, the fitted 1-step detector cannot separate

genuine within-OD routing differences from chance on this behavioural proxy, at any OD-zone granularity tested.

8 Discussion

8.1 Cross-tier read: what generalises

The core inverse-MC scoring machinery ports cleanly across all three tiers: the inverter, FD-checked gradient, contrast diagnostic, and feature plumbing operate unchanged on a synthetic $n = 100$ chain, a SUMO $n = 2,405$ network, and a real-world $n = 1,733$ Xuancheng network. What differs by tier is the loader, the regime definition, and the OD-control logic; the shared scoring core is selected by a single `-dataset {sumo,xuancheng}` CLI flag. This is a useful generalisation result for an inverse-MC pipeline that the original paper validated only on synthetic data and one real-world demonstration.

What does *not* generalise as cleanly is the magnitude of the detection signal. Tier 1 reproduces the paper’s RMAE within ~ 0.05 for $|X_o| \geq 10$. Tier 2 reaches $\text{AUC} = 0.708$ at $T = 40$ on the controlled SUMO comparison, $\sim 69\%$ of the gap to the full-observation empirical 1-step reference of 0.801 , with per-row chain Pearson $\sim +0.21$. Tier 3 on Xuancheng, with no ground-truth sat-navigation label, gives $\text{AUC} \sim 0.53$ on the rush-vs-off-peak behavioural proxy. The drop between Tier 2 and Tier 3 has two distinct contributions:

- The Xuancheng ceiling itself ($\text{AUC}_{\text{emp}} \sim 0.57$) is much lower than the SUMO ceiling (~ 0.69), reflecting that the behavioural rush-vs-off-peak proxy is a weaker contrast than the SUMO labelled comparison.
- The fitted-vs-ceiling gap is similar in fractional terms ($\sim 49\%$ gap closure on pooled Xuancheng vs. $\sim 78\%$ on SUMO at the same $T = 20$ equivalent), so the framework recovers a similar fraction of whatever signal is available — there is just less signal to recover.

The cleanest single-sentence summary is: *under controlled labelled simulation the method recovers a robust fraction of the per-row chain contrast, but on a real-world behavioural proxy the recoverable within-OD contrast, once population-mix is controlled, is small enough to be near chance under this proxy, model class, and data split.*

Table 12 consolidates the three tiers on a common axis: the same inverter and diagnostic produce every row, and the trend in the empirical ceiling — from a noise-free synthetic recovery, through a labelled simulated comparison, to a behavioural real-world proxy — quantifies how much detectable signal each setting actually contains.

Table 12: Cross-tier synthesis of the three validation tiers. The shared inverse-MC scoring core of section 8.1 produces every row; the loader, regime definition, and OD-control logic differ by tier. Tier 2 is the representative seed 42 at $T = 20$ (table 3); Tier 3 is the pooled Apr 17–21 pool at $T = 10$ (table 10). Fitted and Pearson cells are **none/real**. [†]Tier 1 is a π -recovery reproduction scored by RMAE, not a detection AUC: best RMAE 0.19 at $|X_o| = 70$, within ~ 0.05 of the paper’s Fig. 2A (table 2). [‡]The Tier-3 ceiling shown is the un-matched 5-day pool. OD-matching at $K = 8$ lowers it to 0.518 , but a granularity sweep $K \in \{8, 16, 32, 64\}$ returns the matched reference to ~ 0.55 – 0.58 for $K \geq 16$, so no fixed OD-mix fraction is claimed; the fitted detector stays at chance across all K (table 11). The Tier-2 Pearson lift is robust across five seeds ($\Delta = +0.050 \pm 0.023$, $p \approx 0.008$, section 6.3).

tier	dataset (n)	AUC_{emp} ceiling	AUC_{fit}	per-row Pearson
1 synthetic [†]	random graph (100)	—	—	—
2 SUMO	Munich bbox (2,405)	0.685	0.612/0.644	+0.132/ +0.215
3 Xuancheng [‡]	AVI (1,733)	0.573	0.524/0.536	+0.096/ +0.072

8.2 Limits of the chosen approach

Three limitations are foregrounded honestly.

Multi-seed Tier 2; single day-pool on Tier 3. The Tier 2 features result has been confirmed across five SUMO seeds (section 6.3), and that confirmation materially revised the single-seed reading: the aggregate detection-AUC lift is *not* statistically robust ($+0.024 \pm 0.026$, reversing on one of five seeds, $p \approx 0.11$), whereas the per-row chain-recovery lift *is* ($+0.050 \pm 0.023$, positive on all five, $p \approx 0.008$). The multi-seed sweep also reproduced the iter-16 bimodal basin structure, confirming the cross-seed AUC variance is L-BFGS basin-selection rather than measurement noise; a follow-up multistart check (section 6.4) further established that this low basin is *not* escapable by random re-initialisation — zero-init is the best optimum found on both low-basin seeds, so the bimodality is a property of the loss landscape, not of the starting point. Tier 3 (Xuancheng), by contrast, remains a single day-pool (Apr 17–21): a multi-pool replication is the analogous outstanding confirmation there, a compute-cost trade-off rather than a methodological barrier.

L-BFGS basin selection on features=real — now resolved. The Xuancheng OD-matched **features=real** rush-chain fit hit final loss 1397, against the **features=none** fit’s loss 4.058 on the same data, and produced the “impossible” $\text{AUC}_{\text{fit}} = 0.530 > \text{AUC}_{\text{emp}} = 0.518$. We ran the multistart variant (section 7.6, $K = 5$): it escaped the high-loss basin (rush loss $1397 \rightarrow 9.186$, validation-selected), and the correctly-fit chains then gave $\text{AUC}_{\text{fit}} = 0.483$ — chance. The ceiling violation was therefore an optimiser artefact, and removing it *strengthens* the Tier-3 negative result: the within-OD routing contrast is not detectable at the 1-step Markov level under this proxy and data split, rather than being masked by a bad optimum. This also makes the SUMO and Xuancheng basin checks complementary — on SUMO zero-init was already the best optimum (section 6.4); on Xuancheng a better optimum existed but only confirmed the negative result — so neither tier’s headline rests on an optimiser accident.

The rush-vs-off-peak proxy confound — bounded in simulation. The Tier 3 proxy bundles two effects that real data cannot separate: sat-navigation adoption (higher at rush hour) and congestion-induced rerouting that affects all drivers when demand rises. The Tier 2 $\rho \times d$ decomposition (section 6.7) addresses this: holding $\rho = 0$ across the demand axis gives an empirical AUC of 0.493 ± 0.018 (within noise of chance), while holding demand at peak across the ρ axis gives 0.780 ± 0.013 . The full proxy diagonal (0.793 ± 0.023) attributes $\sim 96\%$ of its above-chance signal to sat-navigation adoption, with the pure-congestion contribution within noise and a small residual interaction. The decomposition holds across all five SUMO seeds. The remaining caveat is that SUMO’s $\rho = 0$ vehicles follow precomputed shortest-path routes and do not react to congestion en route, so the SUMO pure-congestion AUC is a *lower bound* on the experienced-human-driver confound; even tripling it would not change the qualitative attribution. Real-world verification of this decomposition would require an alternative experienced-driver routing model in SUMO or, ideally, a real-world dataset with sat-navigation-use labels.

The hitting-rate ($f + g$) extension at scale — refined diagnosis. An earlier draft of this work reported the $f + g$ extension as infeasible at SUMO scale on the strength of a 9.5-hour dense fit that did not terminate. A robust diagnosis (section 6.8) separates that single claim into three: (i) the engineering bottleneck is real but more subtle than “replace dense with sparse” —

a FD-verified sparse-LU implementation now exists, but at SUMO scale ($d \approx 8,400$ wide gradient RHS) dense LAPACK GETRS still wins by $\sim 80\times$ via BLAS3 multi-core kernels; (ii) the binding sensitivity is γ , not the solver: at $\gamma = 0.5$ the floor-dominated L_h overwhelms L_f and $f + g$ hurts, while at $\gamma = 0.95$ on a 200-edge Xuancheng subgraph $f + g$ recovers the full-observation empirical reference within sampling noise; (iii) the data-density issue ($\sim 85\%$ of g cells at floor) persists at every scale we tested, so any future $f + g$ scale-up must operate at high γ . Under the simpler f -only loss the framework remains tractable and that is what the practical detector in this work uses. The natural algorithmic next step is Sherman–Morrison amortisation of the $(I - \beta P_T)$ factor across X_o (one factor per L-BFGS iteration instead of $|X_o|$) or the paper’s natural-gradient descent [1] §4.1; both are flagged as future work.

8.3 Social, environmental, technological implications

Transport-policy modelling. If externally-influenced routing constitutes even a few percent of urban traffic flow, demand models that assume revealed-preference inputs reflect intrinsic preferences are systematically misspecified. A method that quantifies and filters the externally-influenced fraction is a methodological prerequisite for any policy intervention (congestion pricing, infrastructure investment, route-recommendation regulation) whose evaluation rests on preference-revealed data.

Privacy and data access. The most-restrictive privacy control among the three real-world AVI datasets we surveyed is also the one that gives the strongest scientific guarantees: the Padua deposit [13] releases only aggregated flow statistics. The least-restrictive is the Ma *et al.* Xuancheng release [3] which publishes joinable per-trip records with anonymised IDs — enabling the per-trajectory detector this work relies on, but at the cost of stronger reidentification surface area. Methodologies that can validate per-trajectory detection from aggregate-only data would shift this trade-off favourably.

Sat-navigation regulation. The current sat-navigation recommendation pipeline is unregulated in most jurisdictions; the welfare effects of system-wide rerouting around congestion are contested. A practical detector that can statistically attribute trip fractions to sat-navigation guidance would underpin both academic welfare analyses and possible regulatory disclosure requirements.

8.4 Future work

1. **OD-cell bootstrap.** A stratified vehicle-level bootstrap confirms the fitted Tier-3 detector is statistically indistinguishable from chance, and an OD-zone granularity sweep $K \in \{8, 16, 32, 64\}$ confirms this holds at every resolution (table 11); the sweep also showed the $K = 8$ “ $\sim 75\%$ OD-mix” reading to be a coarse-zoning artefact. The remaining outstanding extension is an OD-*cell* bootstrap that resamples which cells enter the match and refits per replicate, quantifying cell-selection variance on top of vehicle-sampling variance — a compute-time rather than methodological gap.
2. **Trajectory-level feature augmentation.** The strict 1-step Markov LR detector ignores trip-level features (length, mean speed, edge-class composition, dwell time at congested edges). A logistic-regression layer on top of $\Lambda(\tau)$ that adds those features would lift detection AUC further at the cost of a slightly weaker chain-recovery claim. Worth doing as a clearly-labelled “deployment-oriented” extension.
3. **Sherman–Morrison factor sharing in $f + g$.** The sparse-LU implementation (sec-

tion 6.8) is FD-verified but slower than dense at SUMO scale because the wide gradient RHS $V \in \mathbb{R}^{n \times d}$ benefits from BLAS3 multi-core kernels. The natural amortisation is to factor $(I - \beta P_T)$ *once* per L-BFGS iteration and derive each per-state h_j and ∇h_j via a Sherman–Morrison update (rank-1 in the row of P_T being zeroed out), which converts $|X_o| = 601$ factor + wide-solve pairs into a single factor and a single wide-solve plus $|X_o|$ rank-1 corrections. This is the algorithmic step that bridges the engineering gap exposed in the subgraph experiment.

4. **Natural-gradient or LM optimisation.** Per section 6.8, even with the engineering fix above, the $f + g$ loss surface is under-constrained under floor-dominated g and benefits from manifold-aware steps. The paper’s §4.1 natural-gradient descent or a Levenberg–Marquardt formulation against the residual Jacobian (iter-17 of the project explored this; abandoned at $d = 8,392$ for memory cost) would combine well with Sherman–Morrison and enable an $f + g$ -with-features re-validation at city scale.
5. **OD-conditional fitting on Xuancheng.** The OD-rebalancing extension matches the marginal ($\text{zone}_o, \text{zone}_d$) distribution. A stronger control fits per-cell chains and aggregates the per-cell contrasts; smaller sample sizes per cell but cleaner methodologically. The loader infrastructure required already exists.
6. **Parametric 2-step Morimura extension.** The regularised 2-step empirical of section 6.5 lifts AUC by $+0.100 \pm 0.009$ across five SUMO seeds, decisively above the 1-step ceiling that the features lever could not pass. The natural inverter extension that this empirical bound motivates is a parametric softmax on (x_p, x) contexts. The state-space expansion is significant ($|X|^2 \approx 5.8\text{M}$ pairs) but only $\sim 3,500$ are observed in training, so the parameter count and gradient cost grow as the observed-pair count, not as n^2 . Per-pair softmax parameters with feature-sharing across topologically similar pairs would be the first design to try.
7. **Experienced-driver routing model in SUMO.** The Tier 2 $\rho \times d$ decomposition (section 6.7) bounds the congestion-routing confound from below, but its $\rho = 0$ vehicles follow precomputed shortest-path routes and do not react to congestion en route. A SUMO scenario in which unaided drivers carry coarse local congestion knowledge (e.g. time-of-day-conditioned edge attractiveness) would tighten the bound to a quantity that more closely tracks human experienced-driver behaviour. SUMO’s routing infrastructure supports this; an off-the-shelf alternative is **rerouting-with-static-traffic-class** priors keyed to rush-hour edge usage.

9 Conclusion

This dissertation designed, validated, and stress-tested a *partial-observation routing-influence filter*: a pipeline that ingests a road graph, regime-labelled partial observations at a sparse subset of stations X_o , and trajectories, and emits a per-trip log-likelihood-ratio score, a thresholded filter decision, and a suite of trust diagnostics (Algorithm 1, fig. 1). The inverse-Markov-chain method of (author?) [1] — originally a single-chain recovery procedure — sits inside the filter as the estimator; the three experimental tiers exist to justify, parameterise, and audit specific filter design choices using a single shared scoring core.

What was designed. A filter with (i) an FD-verified inverse-Markov-chain estimator, including the omega-global gradients that the paper states but does not derive; (ii) a feature-regularised filter variant that wires SUMO-derived edge attributes (road class, speed, lane count, turn

geometry) into the estimator as 10 shared global parameters; (iii) a per-trajectory log-likelihood-ratio scoring rule with a deployment-tunable decision threshold η ; (iv) a diagnostic suite comprising a per-row Pearson/cosine contrast against a full-observation empirical reference, an OD-rebalancing pre-step for real-world population-mix control, multistart basin checks, and a fitting-free empirical 2-step memory audit.

How it works. The estimator fits one regime kernel per regime by minimising the regularised log-ratio loss of [1]; the scorer accumulates $\log \hat{p}_B(x_{t+1} | x_t) - \log \hat{p}_A(x_{t+1} | x_t)$ along each trajectory and thresholds the result; the diagnostic suite flags — before any downstream claim is made — whether the AUC reflects per-row chain recovery or aggregate bias, whether OD-mix has been controlled at the chosen granularity, and whether the chosen optimum survives a multistart audit.

What worked. The synthetic tier (section 5) validates the estimator inside the filter: the reproduction matches the published proposed-method curve closely for $|X_o| \geq 10$, with gradients FD-verified at machine precision on non-zero components. Under controlled SUMO simulation, the full filter reaches $\text{AUC} = 0.708$ at $T = 40$ against a full-observation empirical 1-step reference of 0.801, and the feature-regularised filter variant robustly improves per-row chain recovery across a five-seed sweep (stacked Pearson $+0.050 \pm 0.023$, $p \approx 0.008$, positive on all five seeds). The pipeline ports unchanged at its core to a 1,733-edge real-world network.

What failed or was limited. The aggregate detection-AUC lift from features is seed-dependent, reversing on one of five seeds, so the per-row diagnostic re-classifies features as a chain-recovery lever rather than a robust detection lever — this is exactly the kind of design-relevant verdict the audit suite exists to enable. Conversely, a properly regularised empirical 2-step audit (interpolation backoff with count-conditional trust) lifts the AUC by $+0.100 \pm 0.009$ across five SUMO seeds, with every seed winning — identifying higher-order routing memory as a real signal the 1-step softmax family cannot capture, and promoting a parametric 2-step Morimura extension to a justified future direction. The hitting-rate ($f + g$) route does not naively transfer to full-SUMO scale: a sparse-LU rewrite is FD-verified and works, but at $|X_o| \approx 600$ the wide gradient solve still favours dense BLAS3 over sparse per-column solves by $\sim 80\times$, and a Sherman–Morrison amortisation would be the right algorithmic next step. At a sub-network scale where the framework is operationally fine, $f + g$ at the correctly-tuned loss weight ($\gamma \approx 0.95$) lifts AUC by $+0.055$ and matches the full-observation reference on the test subgraph within sampling noise; the $\sim 85\%$ floor-dominated empirical g persists at every scale we tested and is the γ -tuning constraint. On the real AVI data the fitted 1-step filter does not recover the rush-vs-off-peak contrast at any OD-zone granularity tested ($K \in \{8, 16, 32, 64\}$; AUC 0.476–0.513, every bootstrap CI straddling chance), even though the full-observation reference weakly carries it (~ 0.55 – 0.58 for $K \geq 16$); the multistart audit confirms this is a genuine near-chance outcome rather than an optimiser artefact. An OD-mix attribution we initially read off the coarse $K = 8$ match ($\sim 75\%$) did not survive the granularity sweep and is not claimed.

Where the filter is reliable. Under controlled, labelled simulation with shared OD pairs across regimes, the filter recovers a robust fraction of the per-row routing contrast and supplies a calibrated AUC together with audit signals that correctly distinguish chain recovery from

aggregate bias. The Tier-2 SUMO result is the strongest defensible claim that this dissertation makes about the filter as a positive detector.

Where the filter is not reliable. On a real-world behavioural proxy without ground-truth sat-navigation labels, the within-OD routing signal — once OD-mix is controlled — is not detectable at the 1-step Markov level under the tested proxy, model class, and data split. The filter therefore does *not* support real-world sat-navigation detection on the Xuancheng dataset; Tier 3 is reported as a confounding audit, not a positive detection result. The filter’s optional g input is fully tractable at the sub-network scale where its empirical estimate is informative — where it lifts AUC and recovers the empirical reference within sampling noise — but does not currently transfer to a full city network under naive sparse-LU; a Sherman–Morrison factor-sharing update (or the paper’s natural-gradient descent) is the algorithmic prerequisite for re-enabling the $f + g$ mode at city scale.

Contribution. The contribution is a designed and audited filter pipeline, not a fresh estimator. Academically, the work extends the Morimura framework into a trajectory-level regime detector and supplies diagnostics that distinguish chain recovery from aggregate bias, along with a controlled-simulator decomposition (section 6.7) that bounds the sat-navigation-vs.-congestion confound in the rush-vs-off-peak behavioural proxy used on real-world data, documenting where the approach succeeds (controlled simulation with labels) and where it does not (real-world proxy without labels, OD-mix confounding, 1-step model class, $f + g$ engineering gap at city scale). Practically, it shows that quantifying externally-influenced routing on real urban data is gated less by the inverse-MC machinery than by the availability of ground-truth adoption labels and by control for OD-mix confounding — a result that directly informs how any future real-world sat-navigation detector should be scoped, evaluated, and audited.

10 Project Management and Reflection

This section addresses the *Project Management* and *Reflection* learning outcomes of the module assessment criteria [4].

10.1 Project structure and milestone management

The project was scoped at the outset as a three-tier validation pipeline (synthetic, simulated, real-world) extending a single source paper [1]. This top-down decomposition served well as a stable organising structure: each tier had clear inputs, outputs, and acceptance criteria, and the natural sequencing (upstream validation locks before downstream tiers depend on it) gave a clean de-risking order.

Internal milestone management was driven by a running `research_logbook.md` file that holds twenty “iter- N ” entries spanning the full project. Each iter documents one experimental cycle: hypothesis, setup, result, honest read, and next action. Two practices that worked particularly well:

- **The auto-memory file as an *epistemic* index.** A separate file (`MEMORY.md` with links to per-finding markdown records) was used for “things I now believe and the evidence for them” as opposed to “things I did and when”. Concretely, the file records statements such as “empirical g on SUMO is sparse to the point of being uninformative for the optimiser” with a pointer to the specific iter that established it. This separation made it possible to quickly distinguish “a working assumption that was right when I made it but might now be stale” from “a finding I have evidence for”.

- **Audit-before-extend.** A guiding rule, learned from iter-9’s retraction of an earlier headline number, was to audit the upstream implementation before extending it. Two specific audits saved time: (i) the gradient finite-difference check at $\sim 10^{-9}$ relative error confirmed the analytic gradient derivation before any sweep was run on features, avoiding the trap of attributing fit behaviour to a buggy derivative; (ii) the iter-18 contrast-correlation diagnostic established that AUC alone is an unreliable signal of chain quality, which materially altered how the Phase 4 result was framed.

Two practices that did not work as well are worth recording explicitly:

- **Underestimating the compute scaling of the $f + g$ loss.** The first $f + g$ sweep at SUMO scale was projected at ~ 1 hour and killed at 9.5 hours. The retry at smaller $|X_o|$ and `maxiter` was projected at ~ 30 minutes and killed at 3 hours. Two iterations were burned before the structural diagnosis (the hitting-rate gradient is $O(|X_o| \cdot n^3)$ and the 84%-floor loss landscape is degenerate at this scale) was reached. In future I would profile a single iteration before committing to a sweep at unfamiliar configuration.
- **Late identification of the OD-mix confound.** The Tier-3 rush-vs-off-peak proxy was scoped before the methodological cost of the OD-mix confound was clearly understood. The OD-rebalancing extension — the methodologically correct fix — was added late in the project, and the matched-pool result it produced (section 7.5) materially changed the Tier-3 reading. Earlier scoping would have caught this; the lesson is that “proxy weakness” should be a structured pre-experiment audit item, not an ad-hoc discussion item.

By the assessment criteria of [4], the strengths I would foreground are: contextual investigation (a critical engagement with three real-world AVI datasets, only one of which is actually usable for per-trajectory work, and an explicit positioning against the source paper and rerouting-detection prior art); validation (three independent validation tiers, each with its own honest caveats and a quantified ceiling); and reflection (this section, and the honest framing of the $f + g$ infeasibility, plus the basin-selection question chased to resolution on both tiers via multistart rather than left as a hand-wave). The weaknesses are: the single day-pool on Tier 3 (no multi-pool replication, in contrast to the five-seed Tier-2 confirmation), and that the headline detection-AUC lift from features proved seed-dependent rather than robust once the sweep was run.

10.2 Personal skill development

The technical skills I leaned on heavily and improved in: deriving and finite-difference-verifying analytic gradients (Eqs. 12, 15, 17–18 of the source paper); writing software with a clean separation between the inverter’s mathematical core (in `morimura.py`, kept faithful to the paper) and the application-specific loaders that surround it (`sumo_to_phase3.py`, `real_data/load_xuancheng.py`); and writing a logbook that records both the successful and the unsuccessful experimental cycles in enough detail that I could critique my own past decisions a month later.

The non-technical skill I would name as the one that most shaped the project’s outcome is *honest framing of negative results*. The project did not produce the headline “real-world detector with a high AUC and clean chain recovery” that the SUMO Tier-2 result would have predicted. Instead it produced: a controlled-validation result (~ 0.71 AUC, 69% gap closure on labelled SUMO), a model-class diagnostic (the empirical 2-step detector does not improve the 1-step reference, weakening the case for a parametric 2-step extension), and an honest real-world result (~ 0.53 AUC on a behavioural proxy without ground-truth labels) that quantifies the gap between simulated and real-world detectability. Each of these is a methodologically useful result, in part because they were framed honestly — but the temptation in week 30 of the project to switch the real-world contrast to a stronger but methodologically incorrect proxy

(holiday-vs-normal) had to be resisted. Resisting that temptation in real time, against the prospect of a higher headline number, is the skill I most want to carry forward.

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A Repository structure and reproducibility

The codebase is organised as follows.

morimura.py The inverter core: `Inverter` class with softmax-parametric chain, analytic gradients (local + globals), λ -cross-validated fit, and the synthetic §6.1 reproduction driver.

congestion_filter.py Tier-2-precursor experiments (Phase 2 of the project’s internal phasing): snapshot-level two-regime synthesis and detection.

per_car_detector.py Tier-2 per-trajectory log-likelihood-ratio detector.

sumo_validation/ The SUMO-side machinery: `sumo_to_phase3.py` (network builder, trip reader, feature extractor), `score_seed_big.py` and `contrast_correlation.py` (scorers, with the `-dataset {sumo,xuancheng}` switch), `two_step_empirical.py` (the diagnostic of section 6.5), and `fd_check_features.py` (the FD gradient gate).

real_data/ The Xuancheng tier: the unmodified `xuancheng.net.xml` from [3], the daily JSON trip files, and `load_xuancheng.py` which provides the loader, the three regime-split helpers, and the OD-rebalancing extension.

papers/ Source PDFs of all cited works.

research_logbook.md The day-by-day project log, twenty iter- N entries.

B Use of generative AI

In accordance with the module’s declaration requirement, generative AI was used in this project as follows. An LLM-based code assistant (Claude) was used as a pair-programming aid throughout the implementation phase: drafting code (with the author’s careful review of every line before commit), writing unit-test scaffolding, deriving analytic gradient formulas alongside the author, and stress-testing arguments before they made it into the writeup.

C Build and reproducibility

The project is reproducible from the repository via the README, with the following minimum environment:

```
# Python 3.14, see project README
python3 -m venv .venv
.venv/bin/python -m pip install numpy scipy matplotlib

# Tier 1: synthetic Morimura reproduction (~20 s)
.venv/bin/python morimura.py

# Tier 2: SUMO experiments (requires SUMO_HOME set)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --seed 42 --features real --T 40 --maxiter 1500

# Tier 3: Xuancheng experiments (requires real_data/ populated)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --dataset xuancheng \
    --day 2023-04-17,2023-04-18,2023-04-19,2023-04-20,2023-04-21 \
    --regime_split rush_offpeak --od_match 8 \
    --features real --T 10 --maxiter 1500
```